AutoCFD5

AI/ML TFG summary

Preliminary analysis

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9th October 2026

The AutoCFD AI/ML challenge

Consistent assessment of learned surrogates for road-car aerodynamics

Workshop objective

Assess predictive capability on realistic road-car geometries and develop practical guidance for aerodynamic modelling.

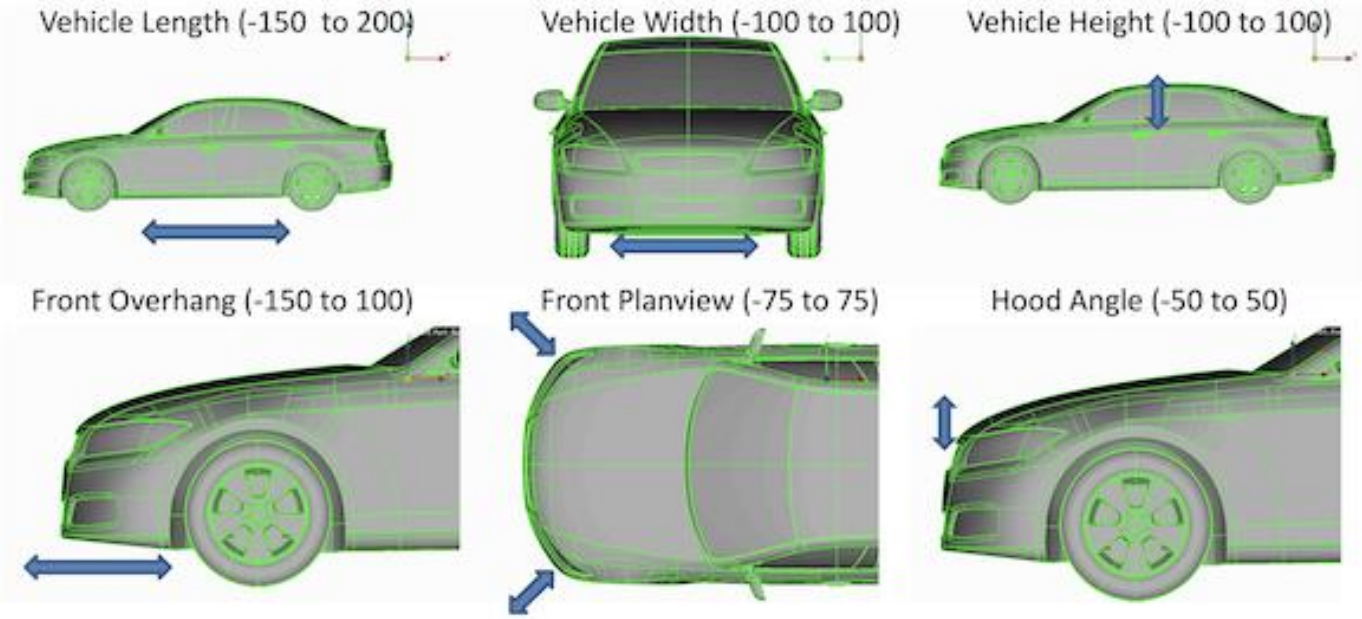
AI/ML focus

Compare surrogate methods using a shared CFD dataset and reproducible evaluation rules.

Assessment question	Evidence
Can the model recover the flow?	Native surface and volume fields
Does it preserve aerodynamic loads?	Drag, lift and pitching moment
Does it generalise?	Reduced-data and extrapolation splits
Where does it fail?	Fixed profiles, Cp cuts and regional errors

DrivAerML dataset

484-geometry design space



Parametrically morphed DrivAer notchback geometry

Consistent scale-resolving CFD provides time-averaged fields and aerodynamic coefficients.

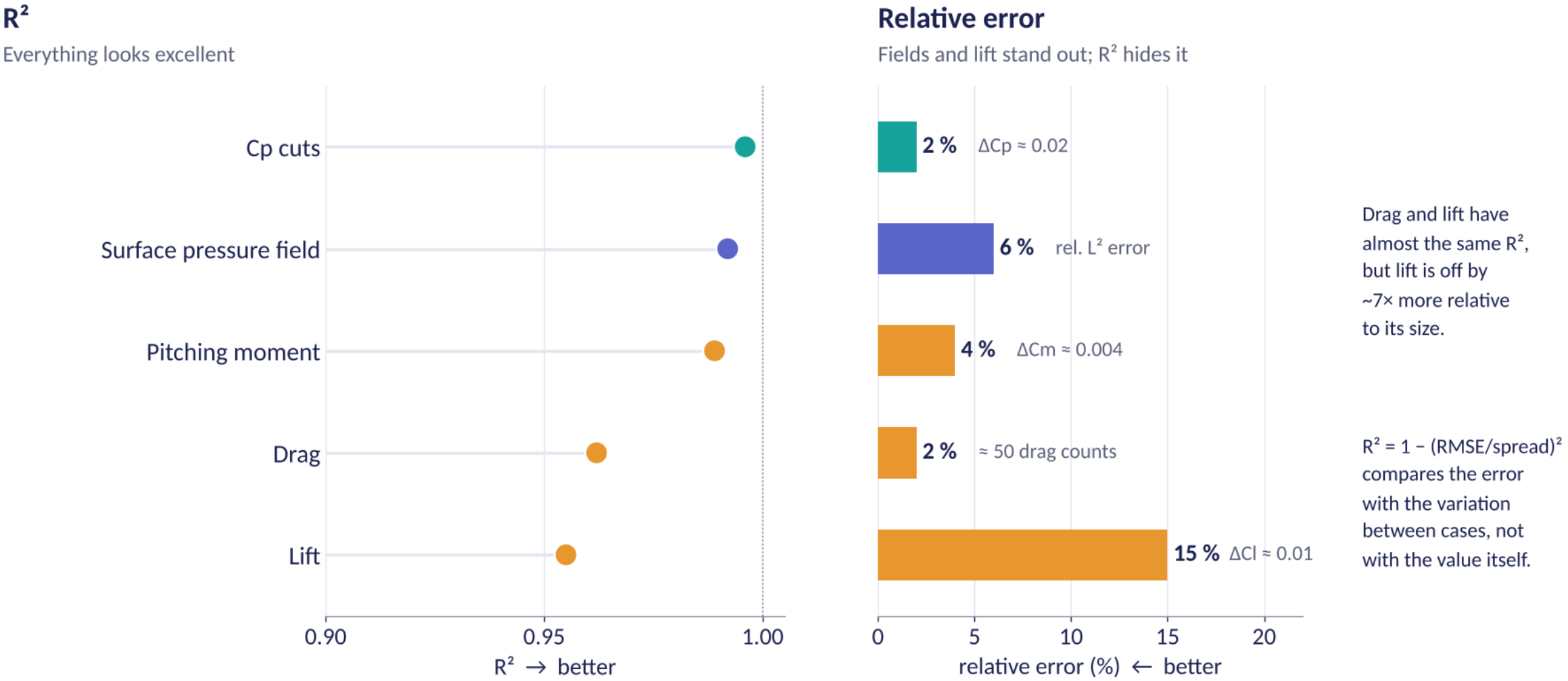
Native evaluation retains the original surface polygons and volume cells.

https://caemldatasets.org/drivaerml/
https://arxiv.org/abs/2408.11969

Eight official evaluation splits

Split	Train	Validation	Test	Purpose
full	400	34	50	Random public baseline
medium	133	34	50	Intermediate data efficiency
scarce	67	34	50	Low-data regime
super scarce	11	34	50	Extreme low-data regime
geometry	339	48	97	Isolated geometry extrapolation
high drag	339	48	97	Highest-drag regime
low drag	339	48	97	Lowest-drag regime
rear separation	339	48	97	Image-derived separation regime

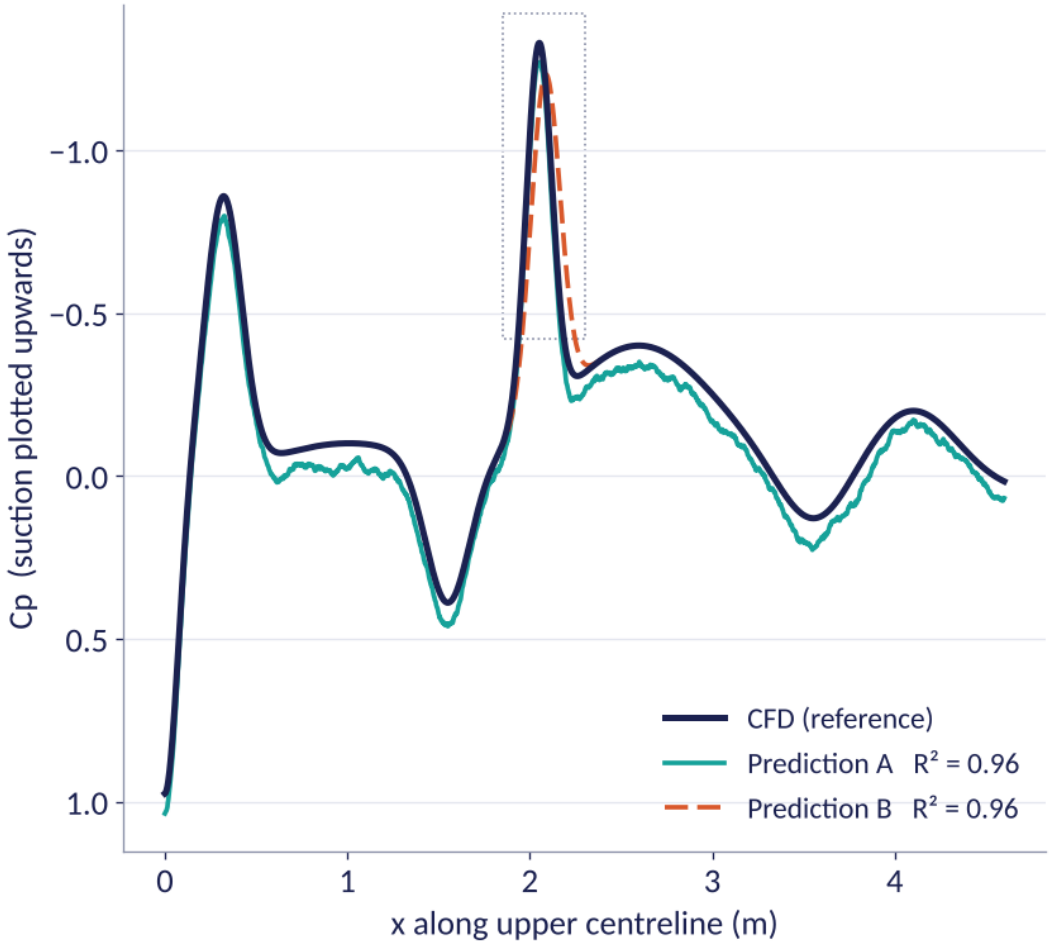
R^2 – One Prediction, many different scores

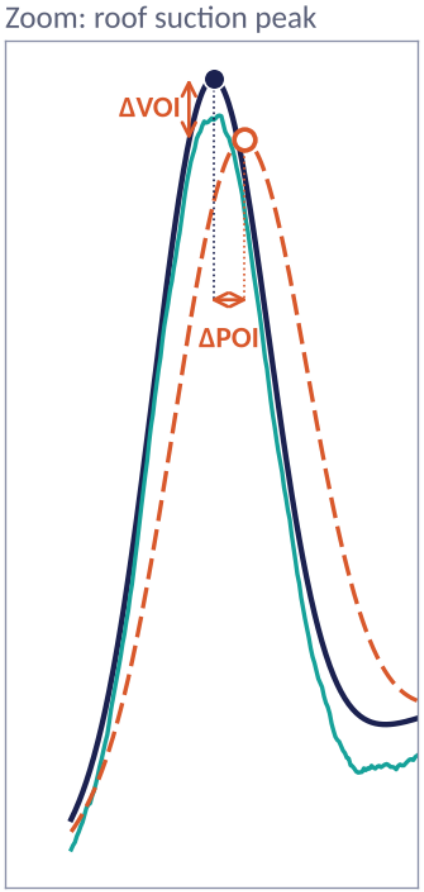


Drag and lift have almost the same R^2, but lift is off by ~7× more relative to its size.

$R^2 = 1 - (\text{RMSE}/\text{spread})^2$ compares the error with the variation between cases, not with the value itself.

Why do we need detailed metrics?



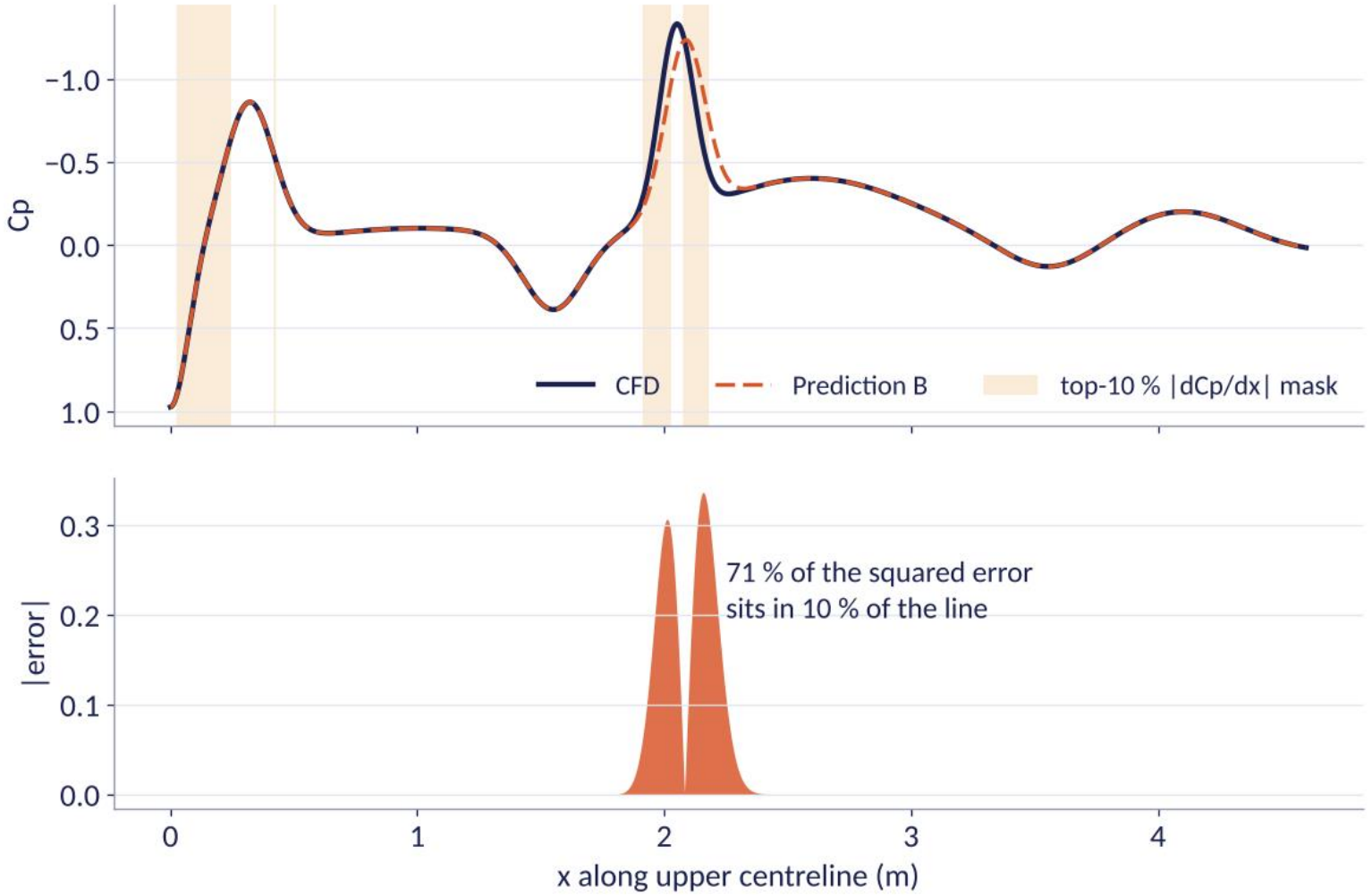


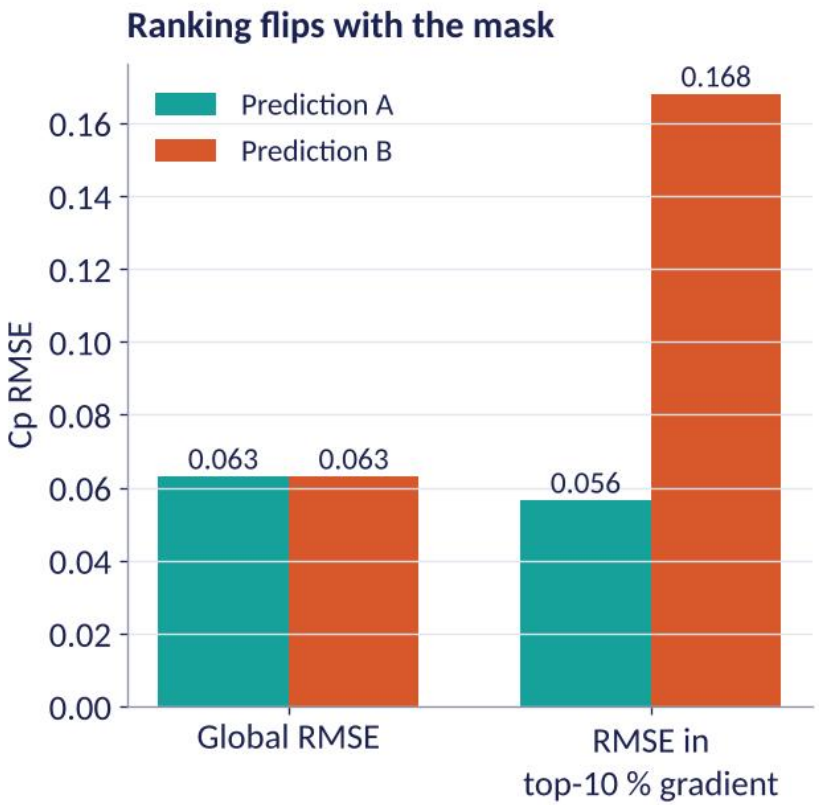
Metric	A	B
Global R^2 (field)	0.961	0.961
Peak position ΔPOI	7 mm	**37 mm**
Peak value ΔVOI	+0.06	**+0.10**
Integrated load error	**+24.3 %**	-0.6 %

A: right shape, offset everywhere → wrong loads

B: exact except at the peak → wrong peak, right loads

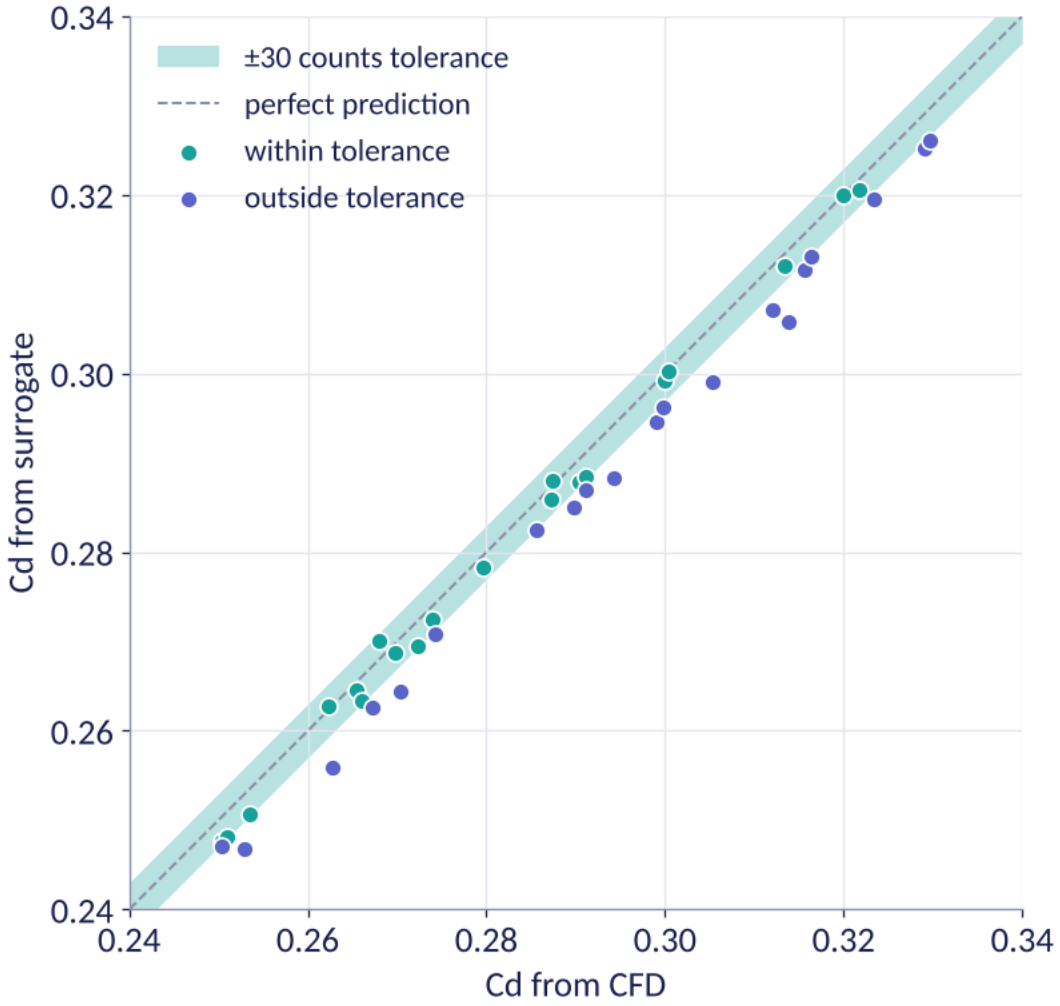
Averaging Effect





Meshes are refined exactly in these regions,
so area weighting shrinks them further.

Translate R^2 into engineering metric



R^2 0.98

looks excellent
in ML terms

37 counts

RMSE, of which
−30 counts is a constant offset

50 %

of cars within
±30 counts

97 %

of car pairs: lower-drag
car identified correctly

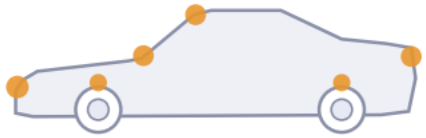
R^2 does not say whether the error is an offset (harmless when comparing variants) or scatter (harmful for ranking them).
It is measured against the car-to-car spread of Cd, so it rewards the overall trend, not the design-relevant increments.

What engineers are interested in

Global
3D field



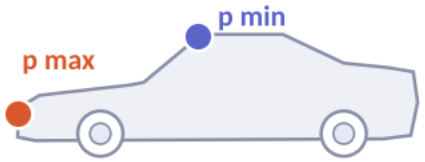
Region
masked field



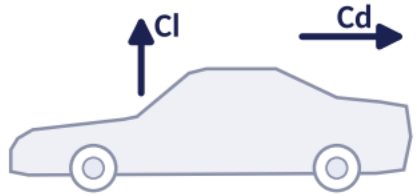
Line
1D cut



Point / value
0D feature



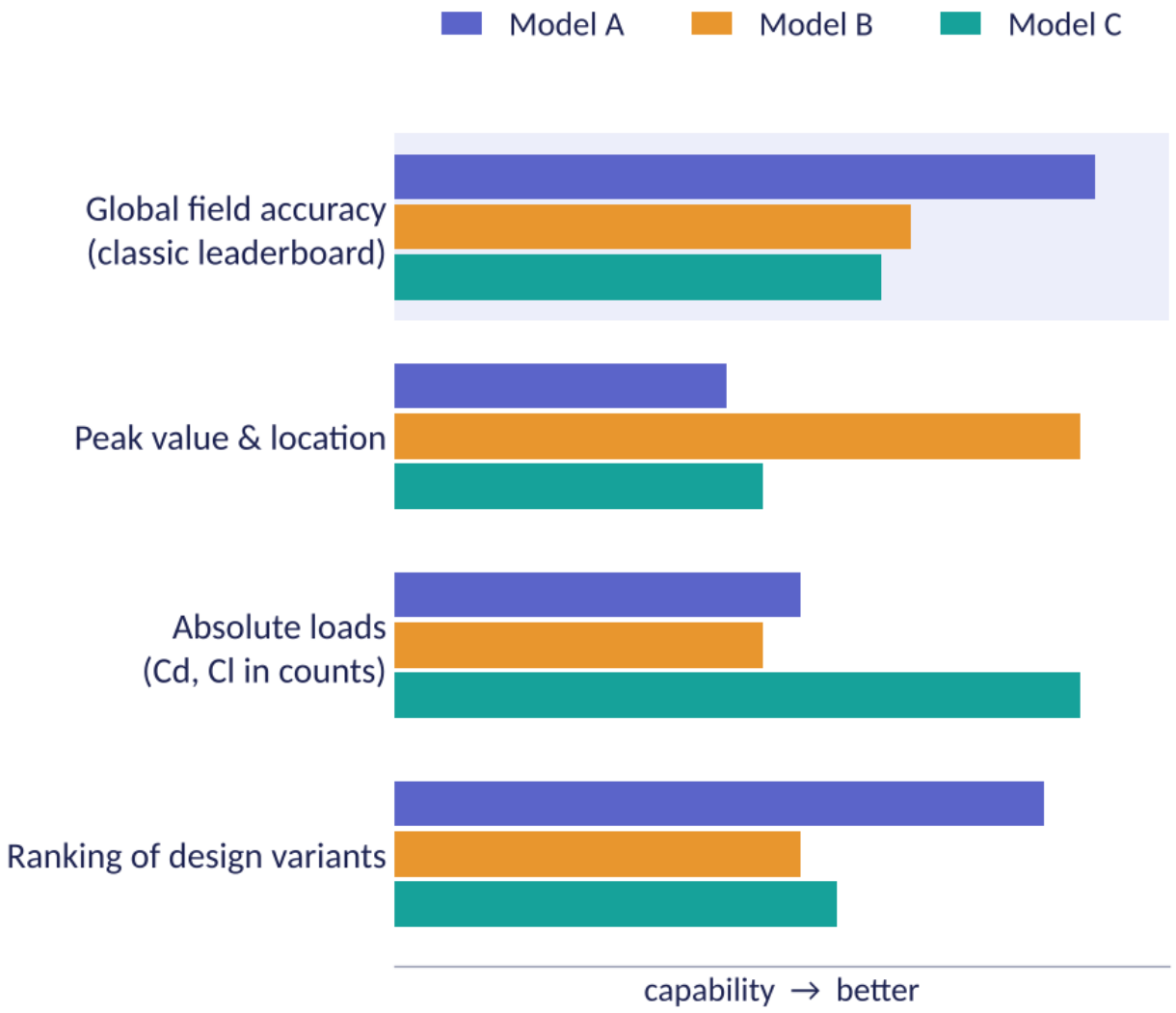
Integral
loads



Category	Engineering question	Example metric
Global	How wrong is the field on average?	area-weighted rel. L^2, RMSE
Region	Is it right where gradients or values peak?	RMSE in top-10 % $\|\nabla p\|$ or $\|p\|$ mask
Line	Does the distribution along a cut match?	Cp-cut, velocity-profile R^2 / RMSE
Point of interest	Is the peak in the right place?	ΔPOI(p_min), ΔPOI(p_max) in mm
Value of interest	Is the peak value right?	ΔVOI(p_min), ΔVOI(p_max)
Integral	Are the loads right, in counts?	Cd, Cl, Cm: R^2, RMSE, ranking

perf. metrics are a diff category

Finding the right metric for your use case



Design exploration

Variants in the design space · needs: ranking

"Is variant 2 better than variant 1?"

best: A

Detail design

Mirror, A-pillar · needs: peaks

"Where is the suction peak, how strong?"

best: B

Vehicle targets

Range, stability · needs: absolute loads

"Is Cd within a few counts?"

best: C

The leaderboard winner (A) is the right choice for only one of three users.

Metrics

Overall score = 50% fields + 25% forces + 25% fixed profiles

Component	Weight	Measure and score transform
Surface pressure / wall shear	15% / 10%	Relative L^2 error, caps 15% / 20%
Volume velocity / pressure	15% / 10%	Relative L^2 error, caps 12% / 15%
Drag / lift / pitching moment	15% / 5% / 5%	R^2 from the declared force route
Fixed velocity profiles / Cp cuts	15% / 10%	R^2 over 16 velocity lines / four Cp cuts

Error component: $100 \times \max(0, 1 - \text{error/cap})$ **R^2 component: $100 \times \text{clip}(R^2, 0, 1)$**

Surface-only ceiling: 60/100. Missing volume fields and velocity profiles receive zero points.

Submission and evaluation workflow

Every model scored on the same native mesh for each test geometry

Participants ran the evaluator locally

01 Export predictions

Declare split and scope. Export complete native surface fields, plus volume fields if submitted.

02 Run the tagged evaluator

Linux / Python environment or supplied container. Run evaluate-entry over the complete test split.

03 Inspect and package results

Review the local report, then run package and verify-package.

04 Deliver a verified ZIP

One compact ZIP per split, with its SHA-256. Organisers verify identities and test-case coverage.

How the comparison stayed consistent

Pinned reference geometry

The dataset revision, native VTP / VTU files and support files were fixed and hash-checked.

Complete native-cell coverage

Every raw_cell_id exactly once. Missing, duplicate or non-finite predictions failed validation. Teams could infer on another representation, then map predictions back to the native cells.

Shared metric definitions

Surface errors used native polygon-area weights; volume errors gave each native cell equal weight. Fixed profile locations, force references and score weights kept the reductions consistent.

Surface-only entries still required the complete surface; unavailable volume components received zero points.

Submission IDs and methods

AutoCFD = 2 ID/submissions , AutoCFD5 = 9/37 ID/submissions

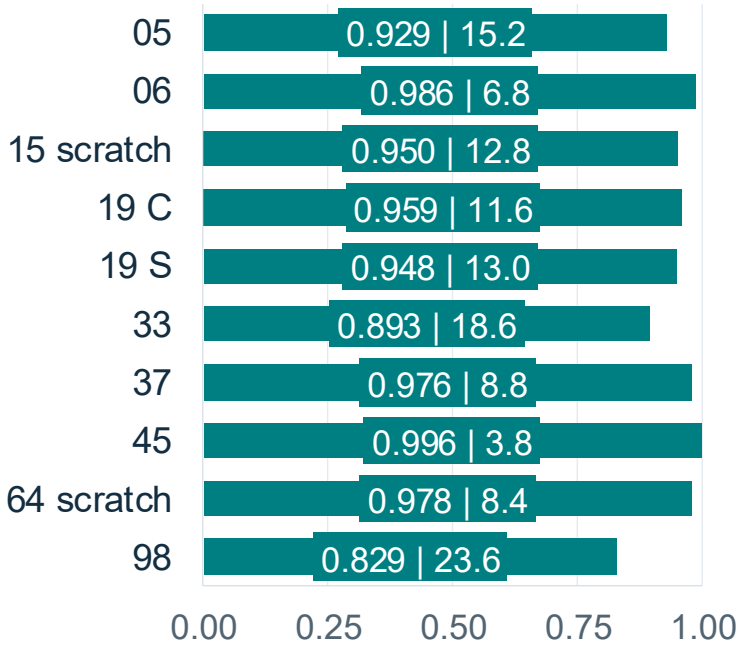
ID	Declared method	Scope	Force route	Packages
45	Transolver	Surface + volume	Integrated	1
37	Transformer-based geometric deep learning	Surface only	Integrated	8
33	Geometric deep-learning	Surface only	Integrated	1
05	Geometric deep-learning	Surface only	Direct	6
98	GeoTransolver	Surface only	Integrated	1
15	Point-cloud based transformer (**scratch + fine-tuned**)	Surface + volume	Integrated	8
64	Geometry-conditioned Transformer (**scratch + fine-tuned**)	Surface + volume	Integrated	5
6	Anchor-based Universal physics transformer (AB-UPT)	Surface + volume	Integrated	1
19	Hybrid attention - graph neural network	Surface only	Integrated	6

Full split: coefficient R^2 and absolute error

All submissions show good force prediction (<10 drag counts)

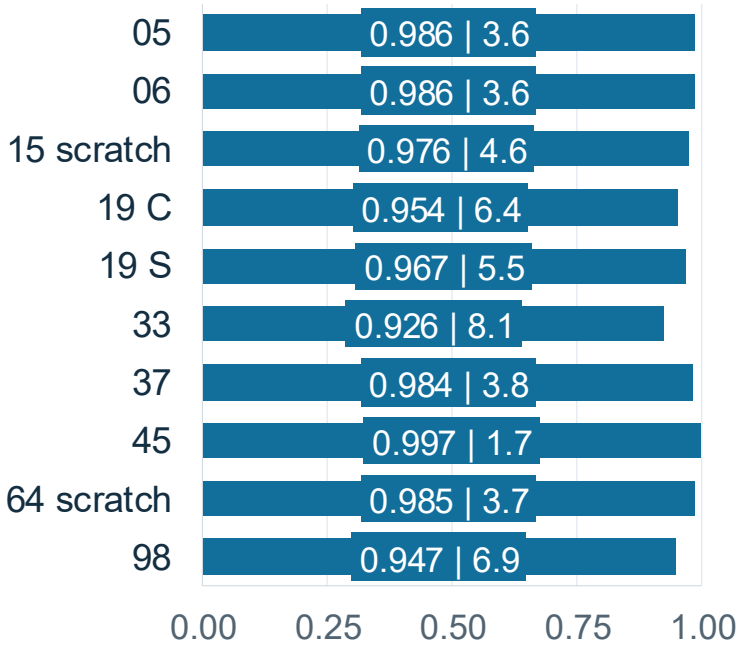
CL — lift

R^2 | RMSE in lift counts



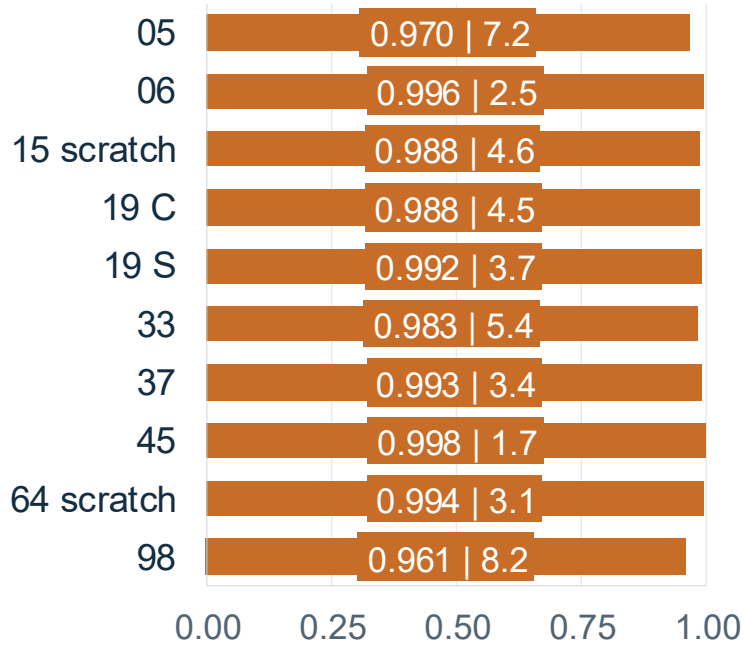
Cd — drag

R^2 | RMSE in drag counts



Cm — pitching moment

R^2 | RMSE in pitching-moment counts



Labels: R^2 | RMSE. Higher R^2 and lower RMSE are better.

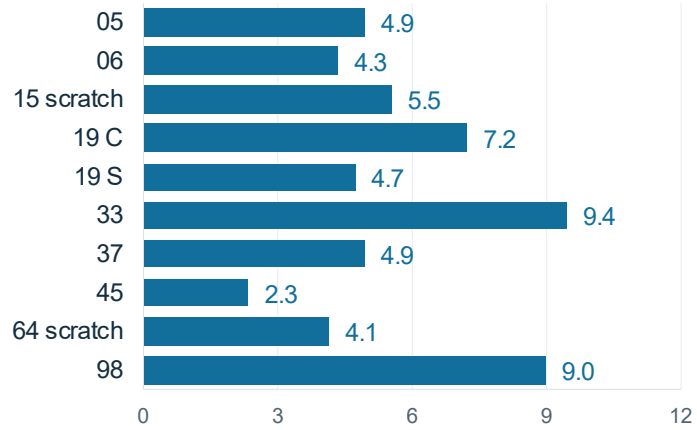
05: direct coefficients; others: integrated fields. 19 C/S: combined/separate.

Full split: geometry-to-geometry prediction

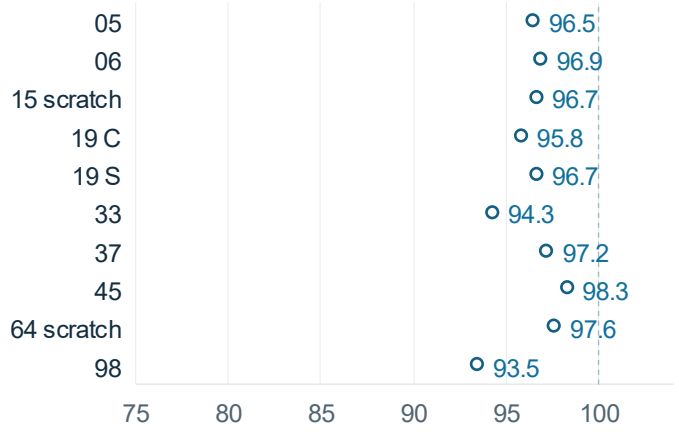
50 test geometries, 1,225 shared pairs.

Cd: drag

Delta RMSE (counts), lower is better

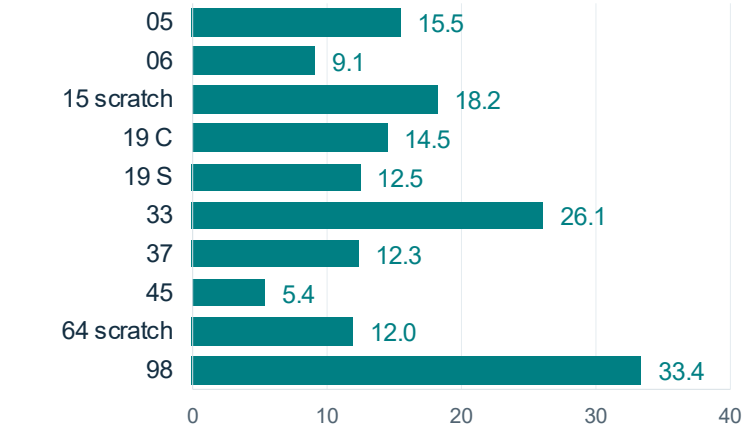


Correct direction (%), higher is better

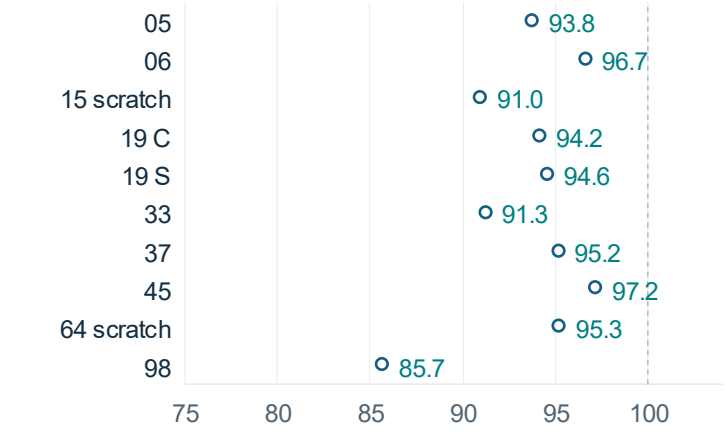


CL: lift

Delta RMSE (counts), lower is better

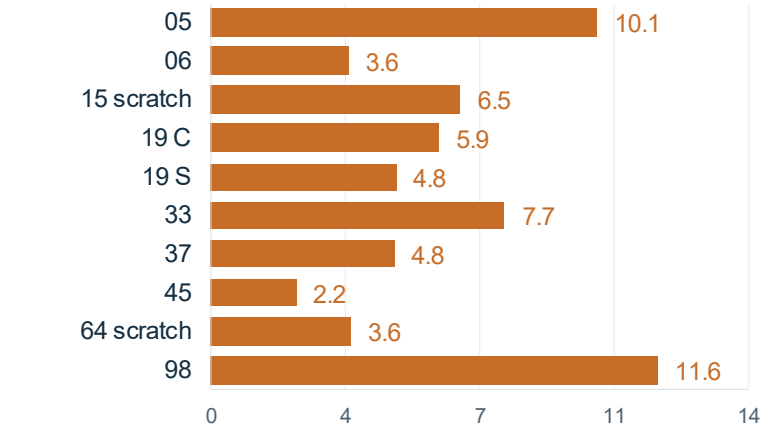


Correct direction (%), higher is better

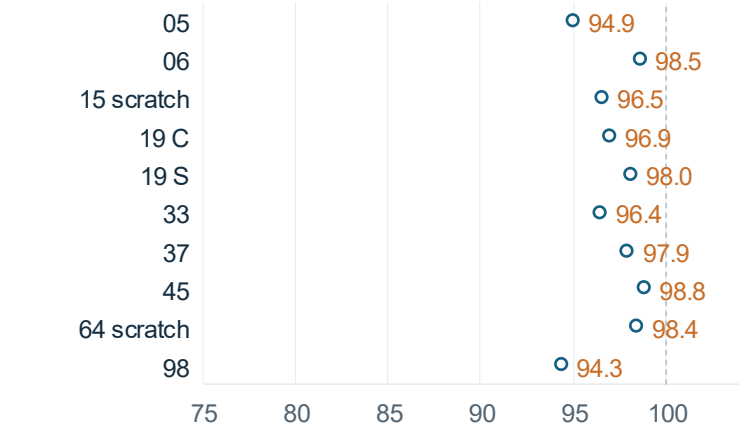


Cm: pitching moment

Delta RMSE (counts), lower is better



Correct direction (%), higher is better



Δ error = [ML(B) − ML(A)] − [CFD(B) − CFD(A)]. 1 automotive count = 0.001 in the respective coefficient.

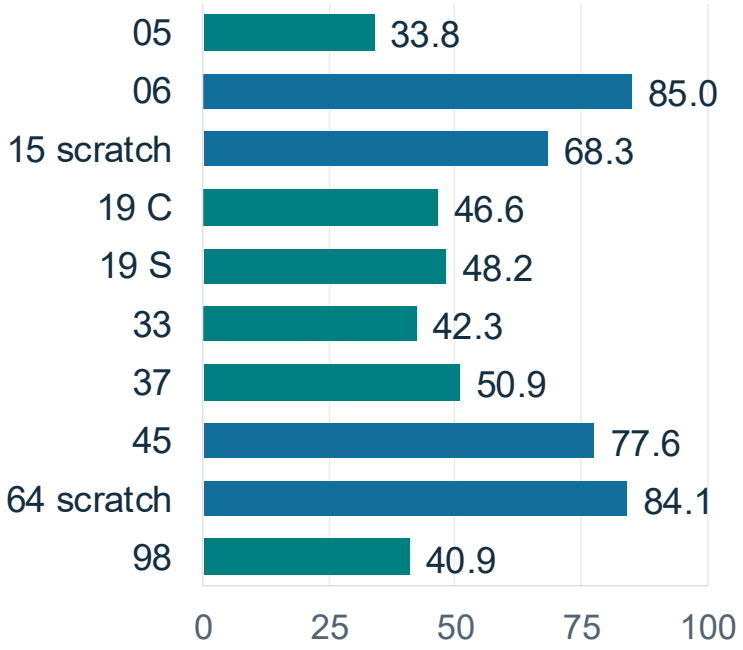
05: direct coefficients, others: integrated fields. 19 C/S: combined/separate.

Full split: score and surface-field errors

Much larger spread across overall score and surface p/wss.
L2 isn't enough to understand differences

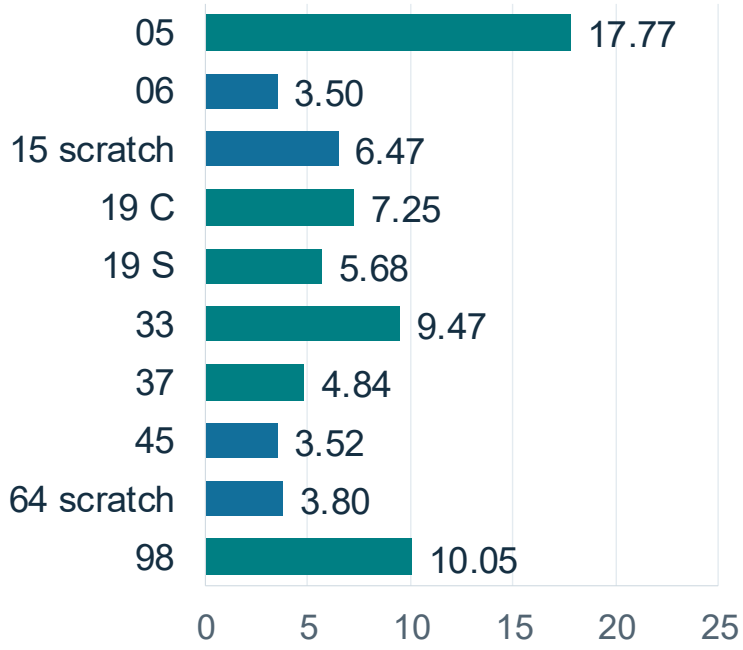
Overall score

Score / 100, higher is better



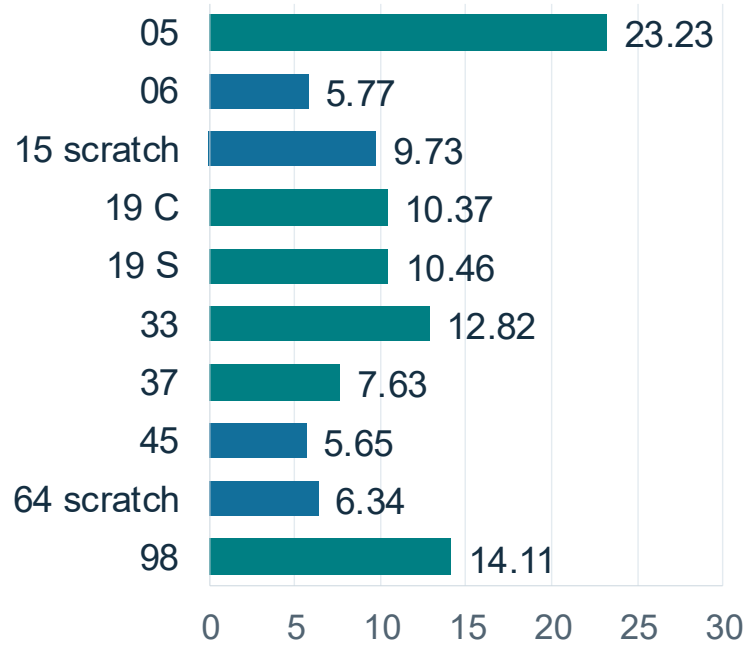
Surface pressure

Relative L² error (%), lower is better



Surface wall shear

Relative L² error (%), lower is better



Teal: surface only, score ceiling 60. Blue: surface + volume, score ceiling 100.

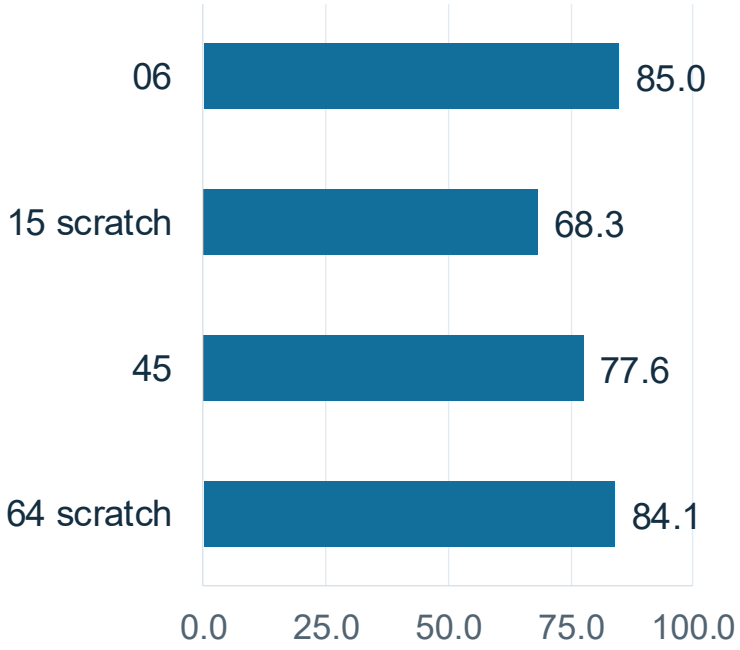
ID 19 C/S: combined/separate.

Full split: score and volume-field errors

400 training cases, same 50 test cases. 4 submission IDs, 4 configurations.

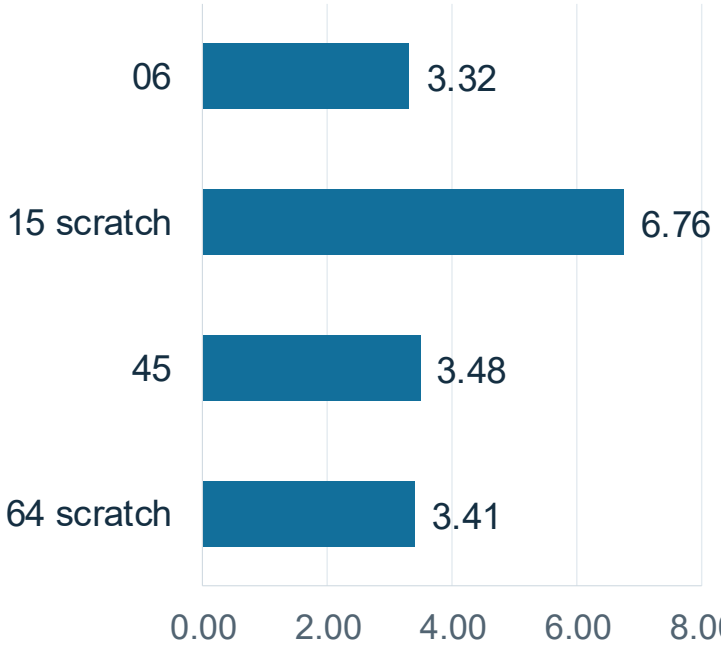
Overall score

Score / 100, higher is better



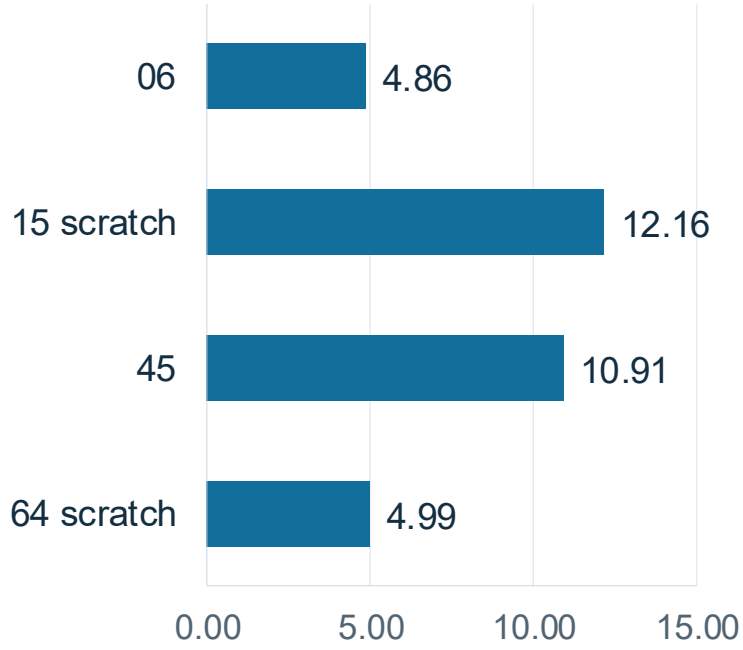
Volume pressure

Relative L² error (%), lower is better



Volume velocity

Relative L² error (%), lower is better



Volume submitters only. Overall score includes forces, surface and volume metrics.

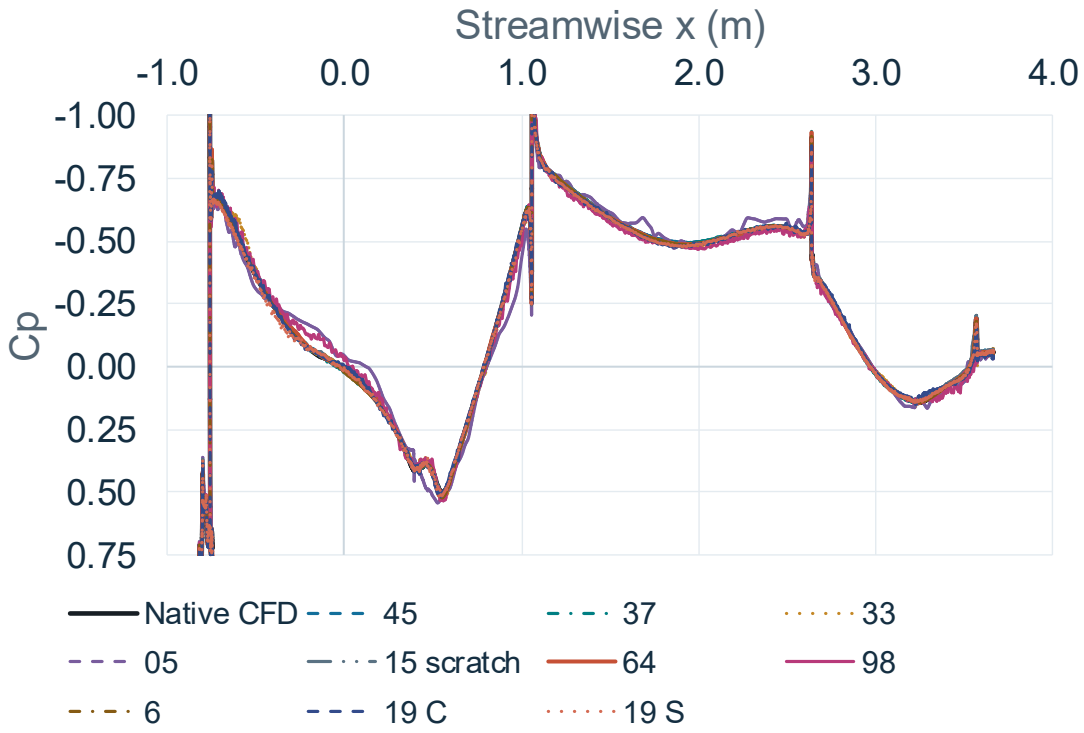
Fine-tuned results excluded, matching the preceding Full-split comparison.

ID 45: legacy evaluator v1.1.3.

Cp comparison: body centrelines (run 215)

Only one outliner (which had higher L2 for p/wss) - 3.4%-10.4% show very similar Cp profile

Upper-body centreline



Underbody centreline

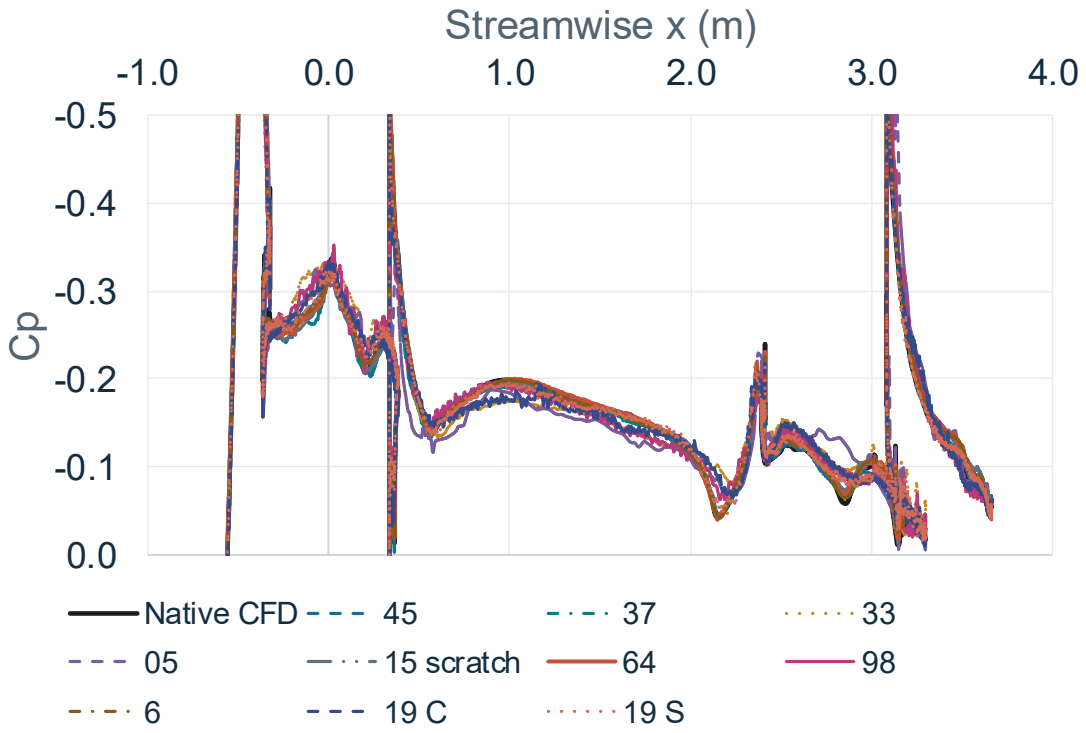


19 C: RWR combined · 19 S: RWR separate. Both surface only. 45: legacy.

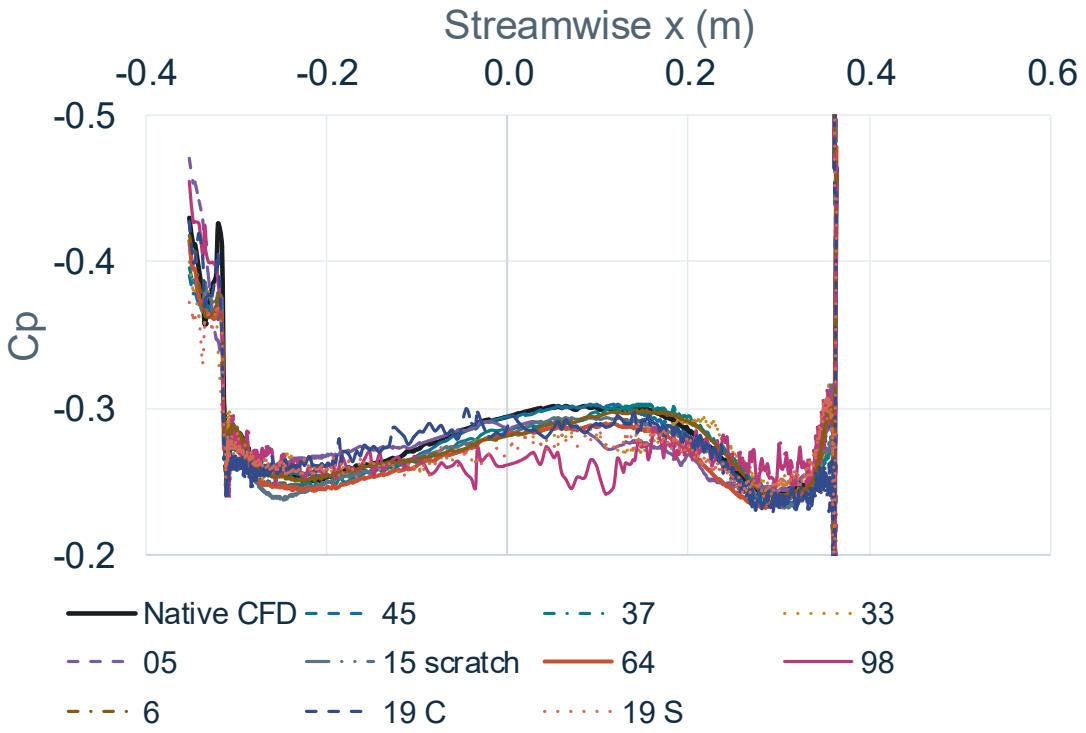
Cp comparison: sidewall and wheelhouse (run 215)

Only one outliner (which had higher L2 for p/wss) - 3.4%-10.4% show very similar Cp profile

Sidewall, z = 0.15 m



Front-left wheelhouse, y = −0.6 m

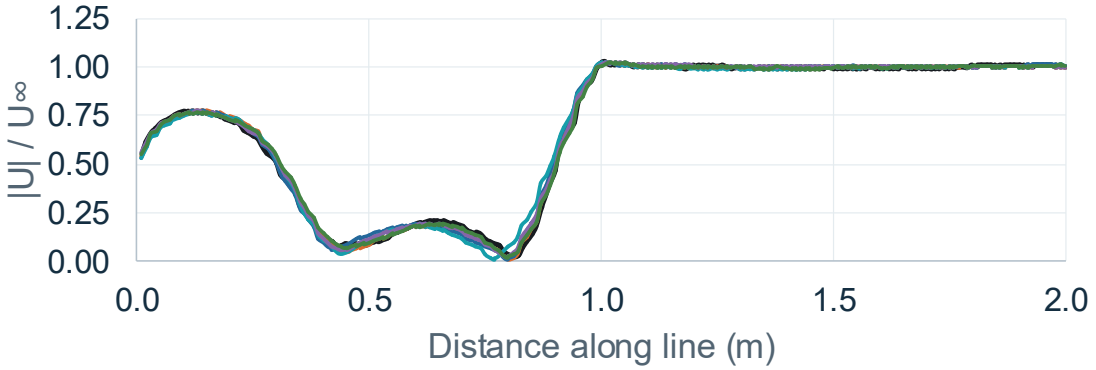


Full split: velocity profiles at four V stations

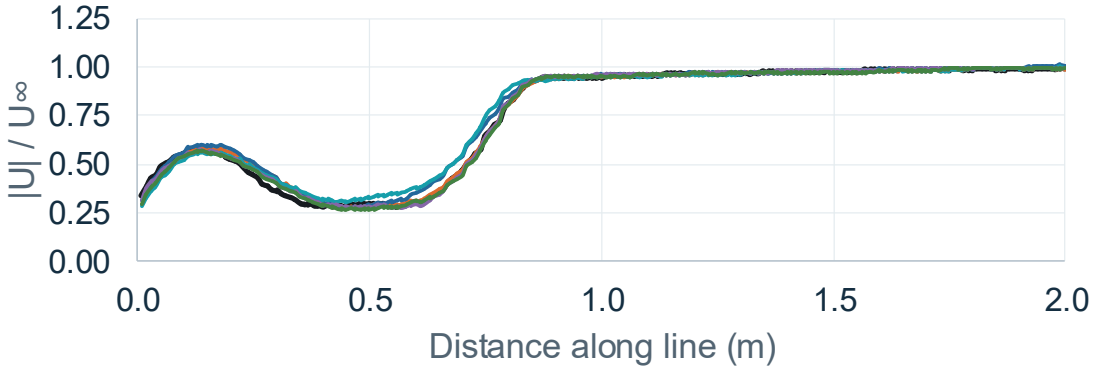
Run 215 · V3–V6, y = 0 m · Five volume configurations

Reference CFD **6 R^2 0.9966** **15 S R^2 0.9943** **15 FT R^2 0.9948** **45 R^2 0.9985** **64 R^2 0.9955**

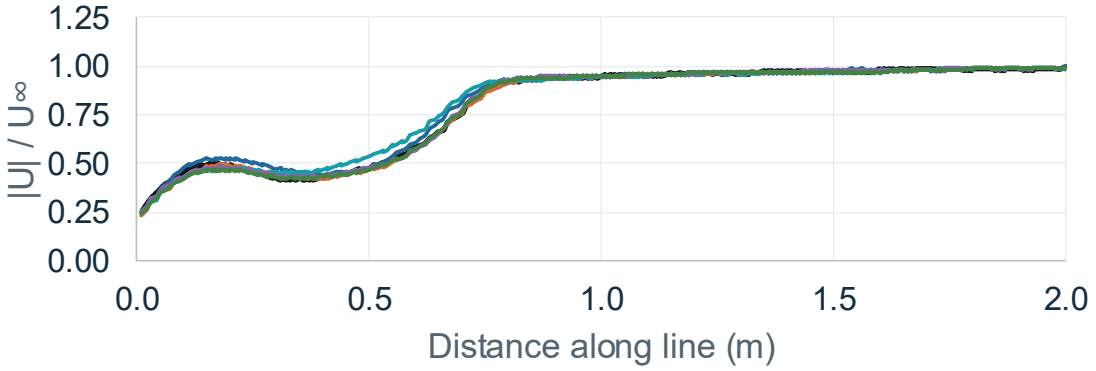
V3 · x = 4.007 m



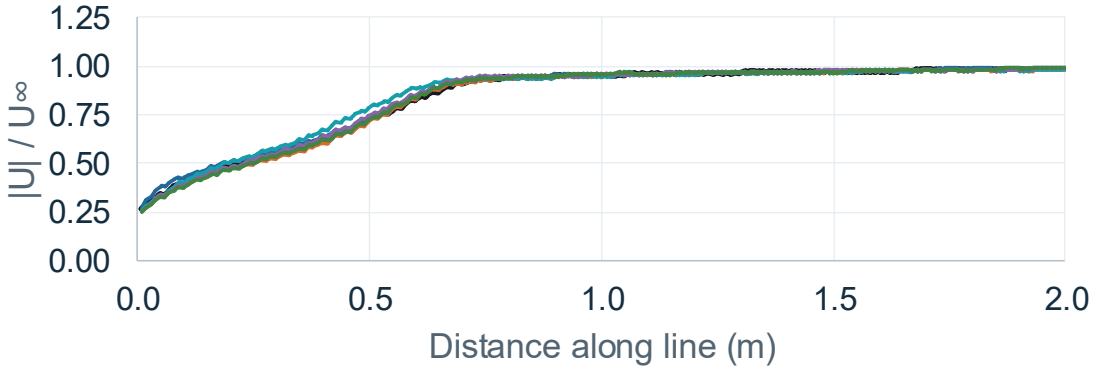
V4 · x = 4.657 m



V5 · x = 5.007 m



V6 · x = 5.507 m



Legend R^2: overall velocity profiles, 16 lines × 50 test cases. Native samples and gaps preserved.

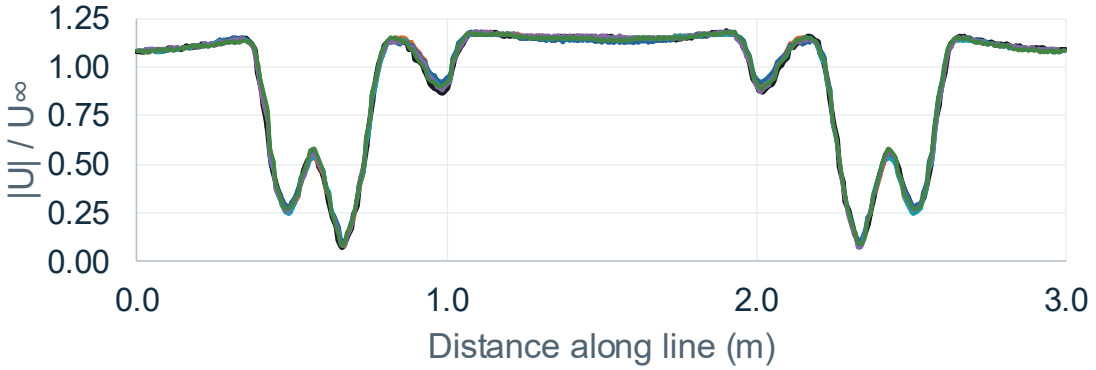
Run 215: near ID 6's median error. 15 S: from scratch. 15 FT: AeroSUV fine-tuning.

Full split: velocity profiles across four U cuts

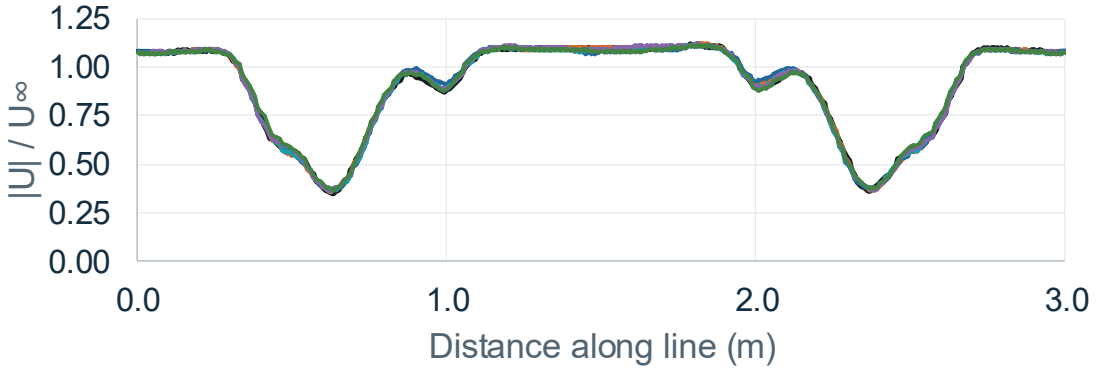
Run 215 · U2–U5, z = −0.2376 m · Five volume configurations

Reference CFD **6 R^2 0.9966** **15 S R^2 0.9943** **15 FT R^2 0.9948** **45 R^2 0.9985** **64 R^2 0.9955**

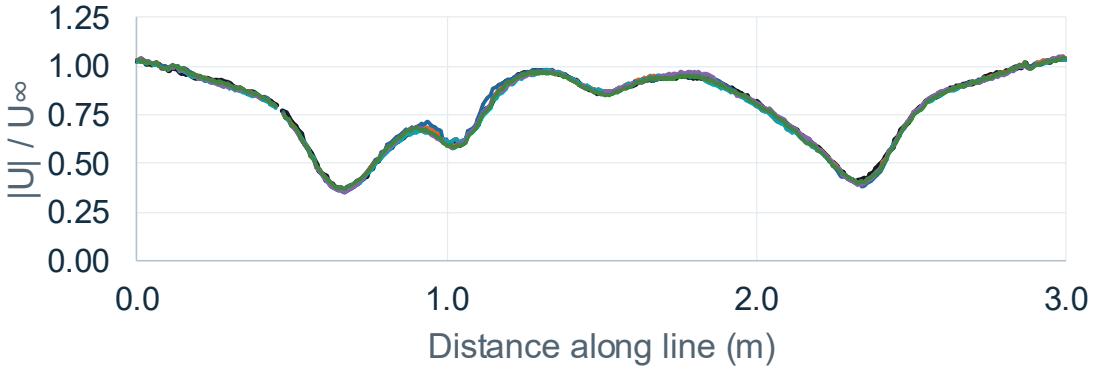
U2 · x = 0.357 m



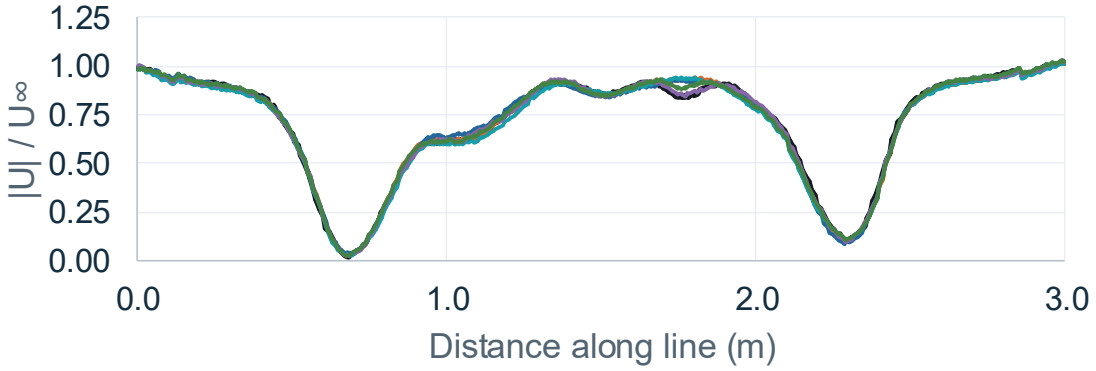
U3 · x = 0.707 m



U4 · x = 2.457 m



U5 · x = 3.157 m



Legend R^2: overall velocity profiles, 16 lines × 50 test cases. Native samples and gaps preserved.

Run 215: near ID 6's median error. 15 S: from scratch. 15 FT: AeroSUV fine-tuning.

Training data volume: velocity at four V stations

ID 15 , from scratch · Run 215 · V3–V6, y = 0 m

Reference CFD

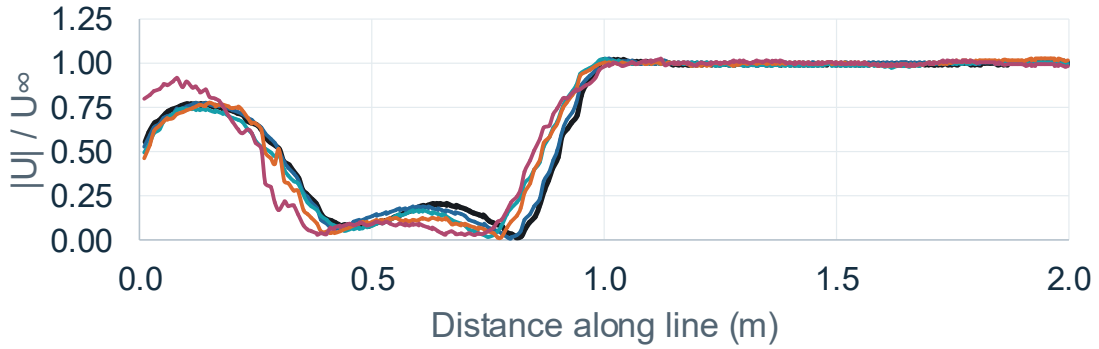
Full (400 train)
Overall R² 0.9943

Medium (133 train)
Overall R² 0.9777

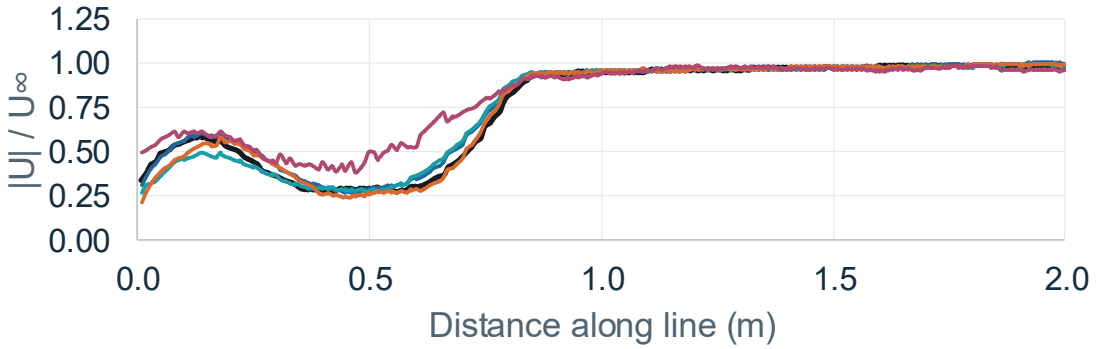
Scarce (67 train)
Overall R² 0.9538

Super-scarce (11 train)
Overall R² 0.8140

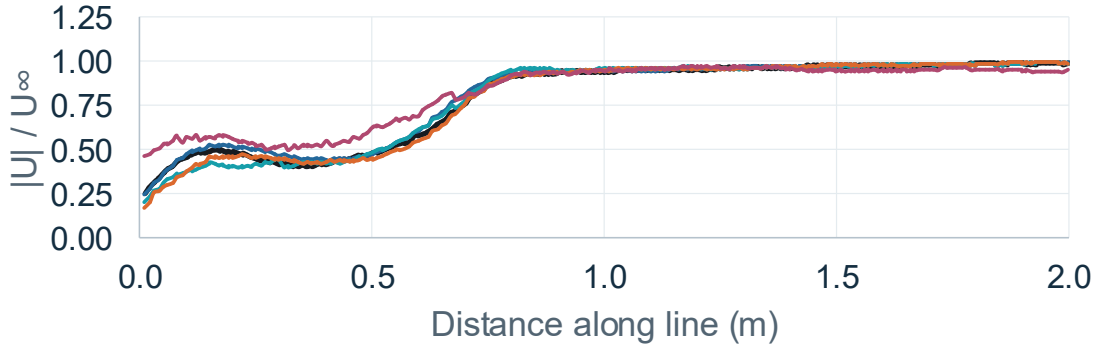
V3 · x = 4.007 m



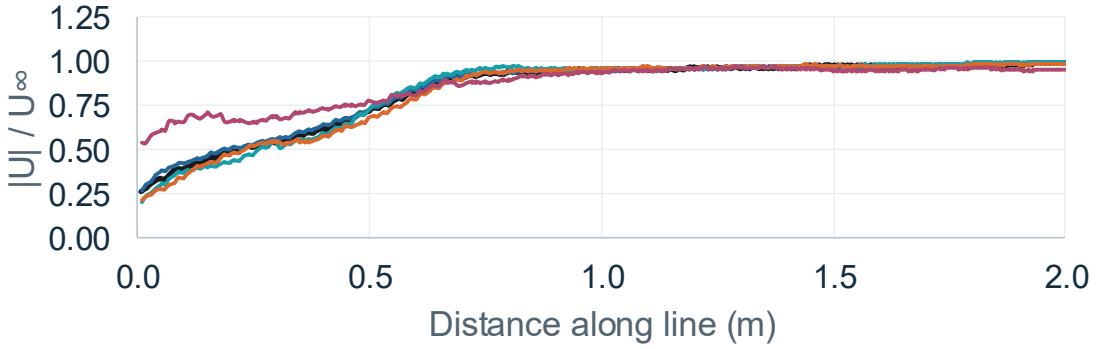
V4 · x = 4.657 m



V5 · x = 5.007 m



V6 · x = 5.507 m



Training data volume: velocity across four U cuts

ID 15, from scratch · Run 215 · U2–U5, z = −0.2376 m

Reference CFD

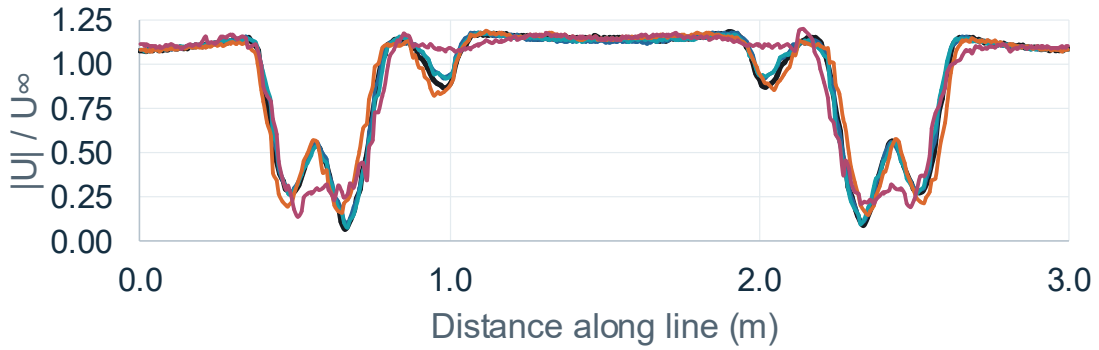
Full (400 train)
Overall R² 0.9943

Medium (133 train)
Overall R² 0.9777

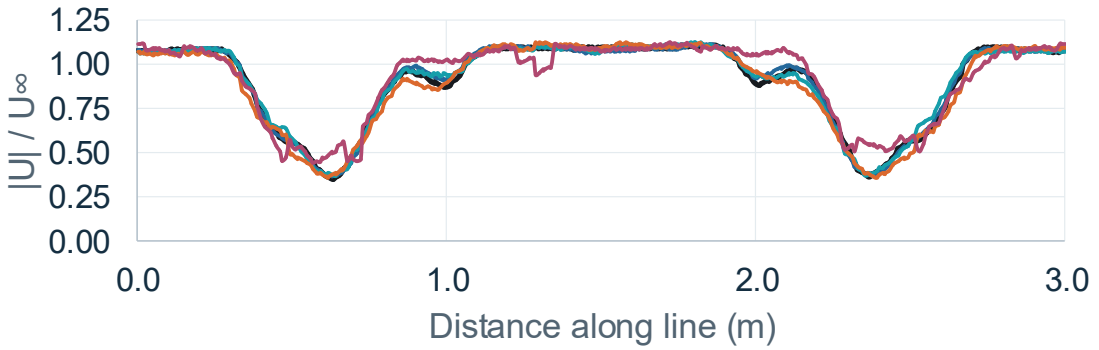
Scarce (67 train)
Overall R² 0.9538

Super-scarce (11 train)
Overall R² 0.8140

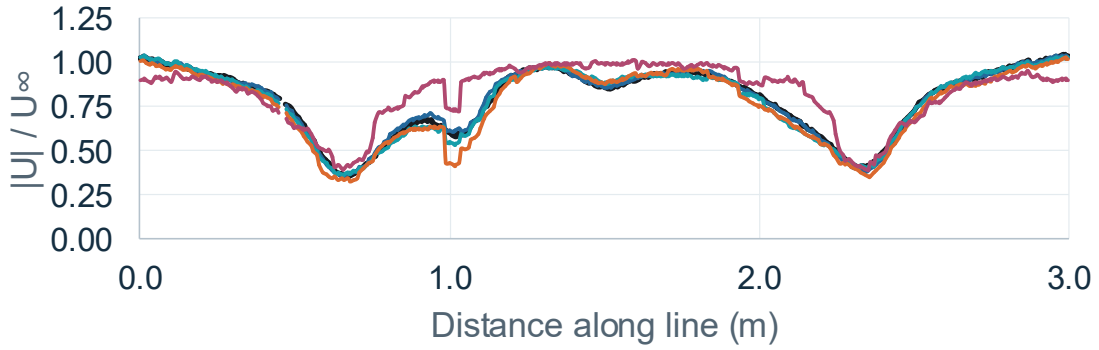
U2 · x = 0.357 m



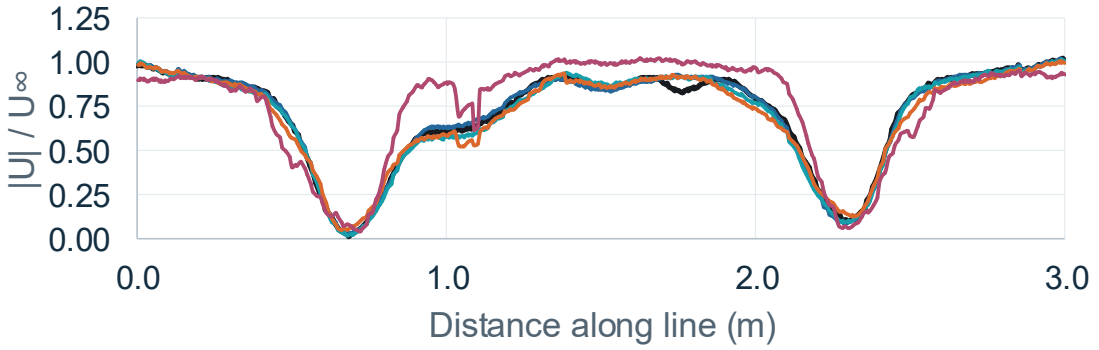
U3 · x = 0.707 m



U4 · x = 2.457 m



U5 · x = 3.157 m



Overall R² covers 16 velocity lines × the same 50 test geometries. Native samples and gaps preserved.

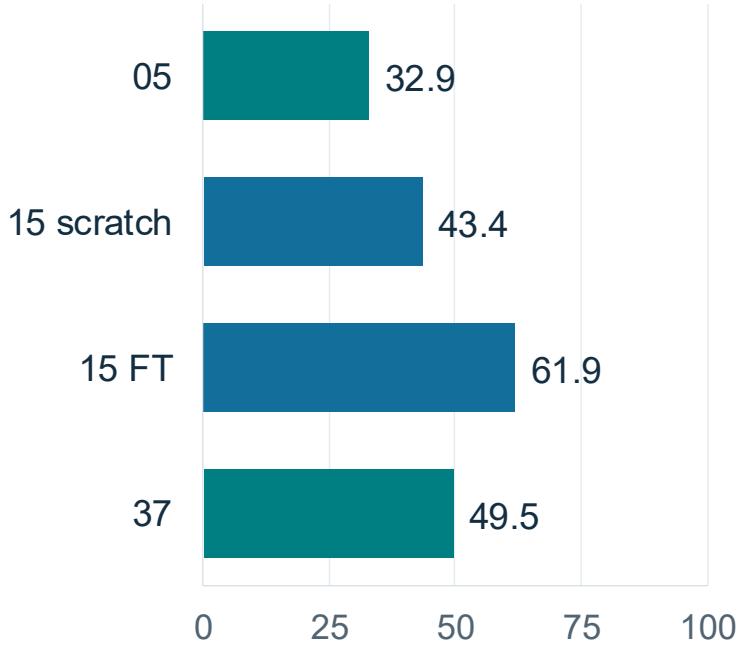
Same model family, from scratch, 500 epochs. Run 215 is one illustrative test geometry.

Medium split: score and surface-field errors

133 training cases, same 50 test cases. 3 submission IDs, 4 configurations.

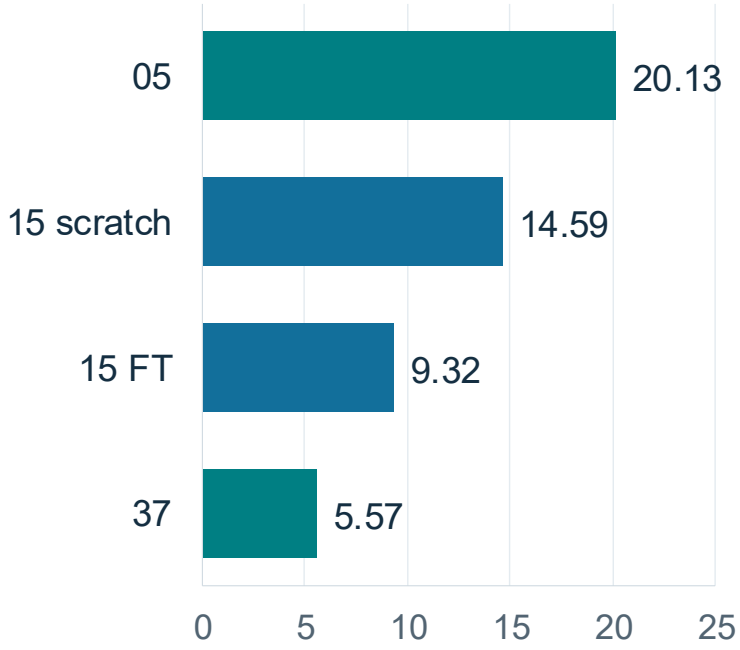
Overall score

Score / 100, higher is better



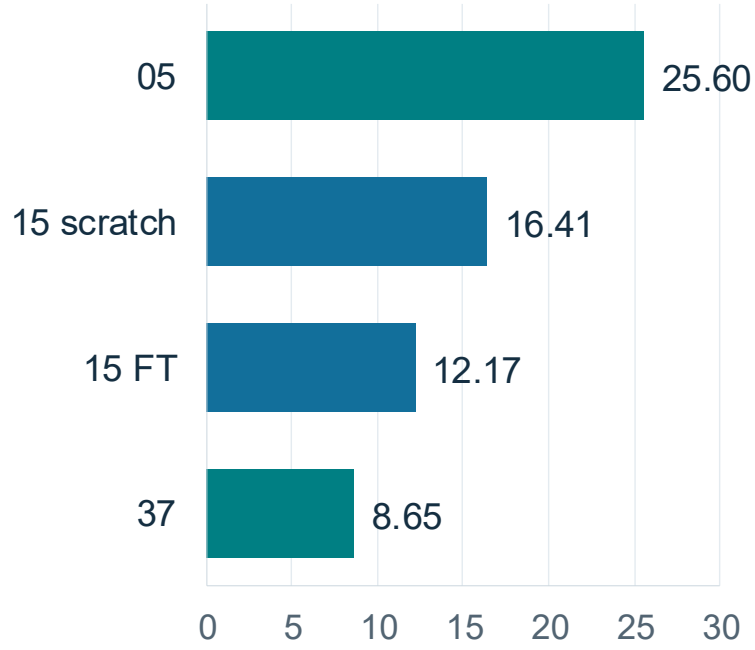
Surface pressure

Relative L² error (%), lower is better



Surface wall shear

Relative L² error (%), lower is better



Teal: surface only, score ceiling 60. Blue: surface + volume, score ceiling 100.

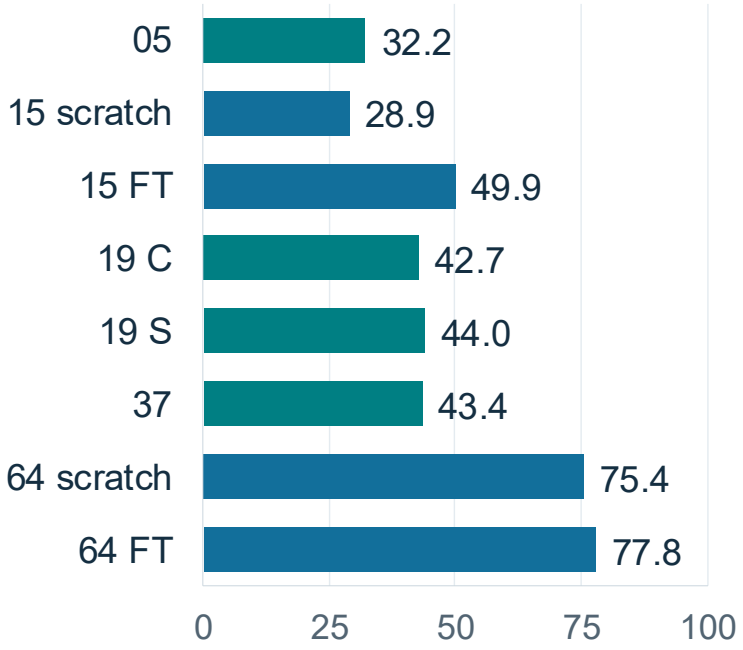
FT: external pretraining + fine-tuning. ID 19 C/S: combined/separate.

Scarce split: score and surface-field errors

67 training cases, same 50 test cases. 5 submission IDs, 8 configurations.

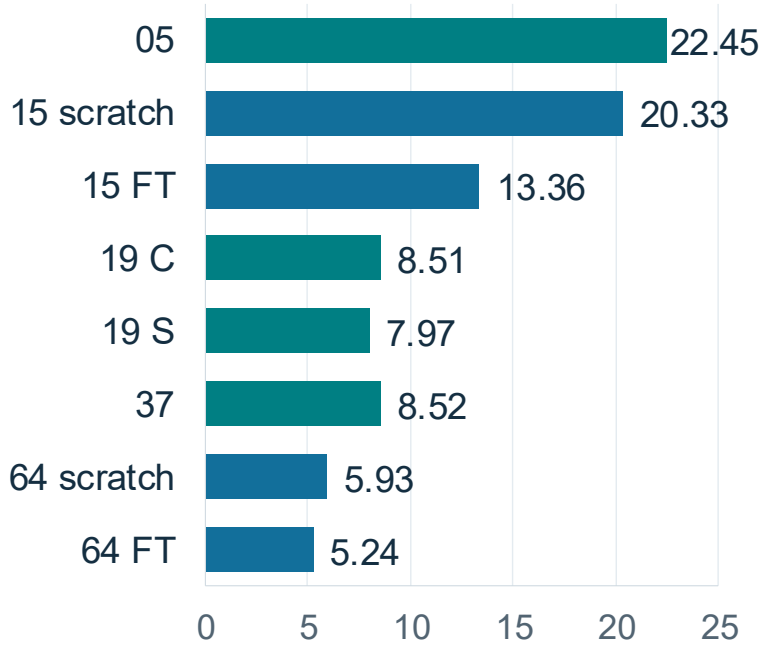
Overall score

Score / 100, higher is better



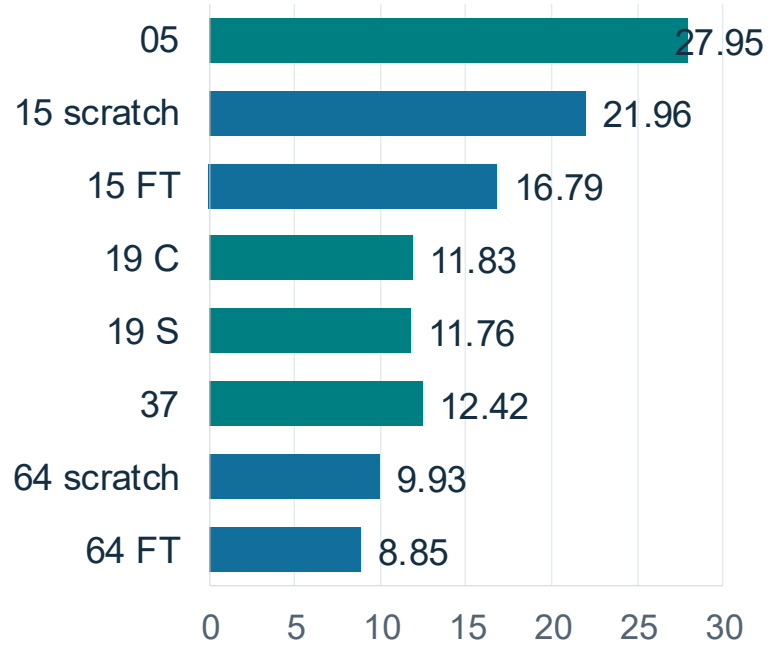
Surface pressure

Relative L² error (%), lower is better



Surface wall shear

Relative L² error (%), lower is better



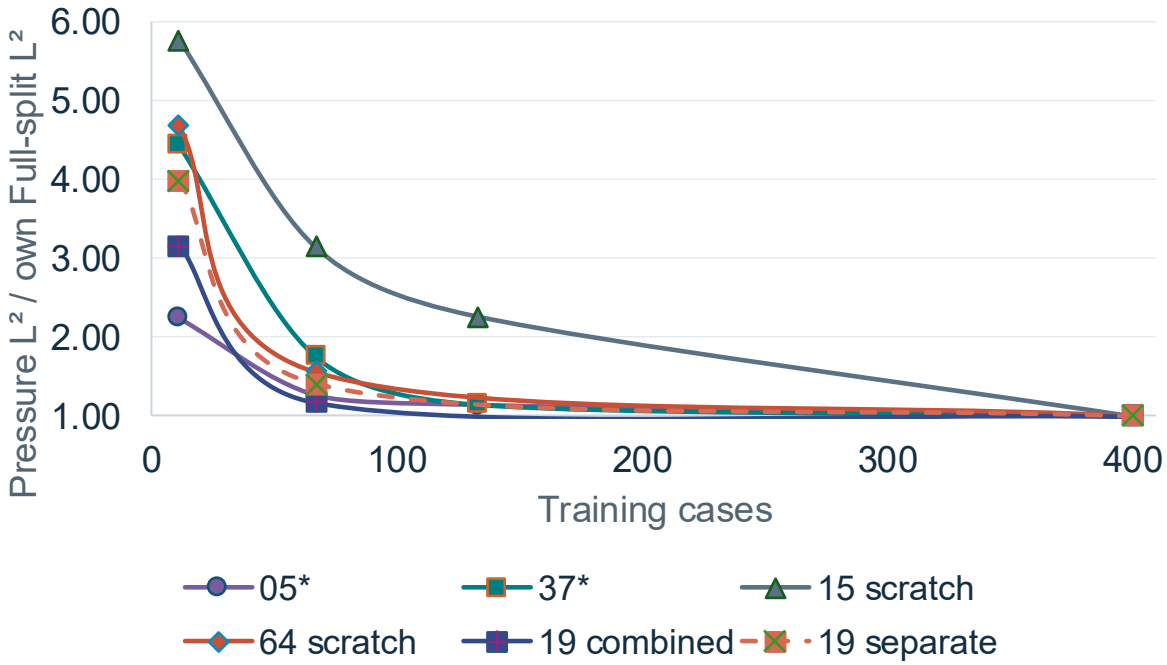
Teal: surface only, score ceiling 60. Blue: surface + volume, score ceiling 100.

FT: external pretraining + fine-tuning. ID 19 C/S: combined/separate.

Training data volume and field accuracy

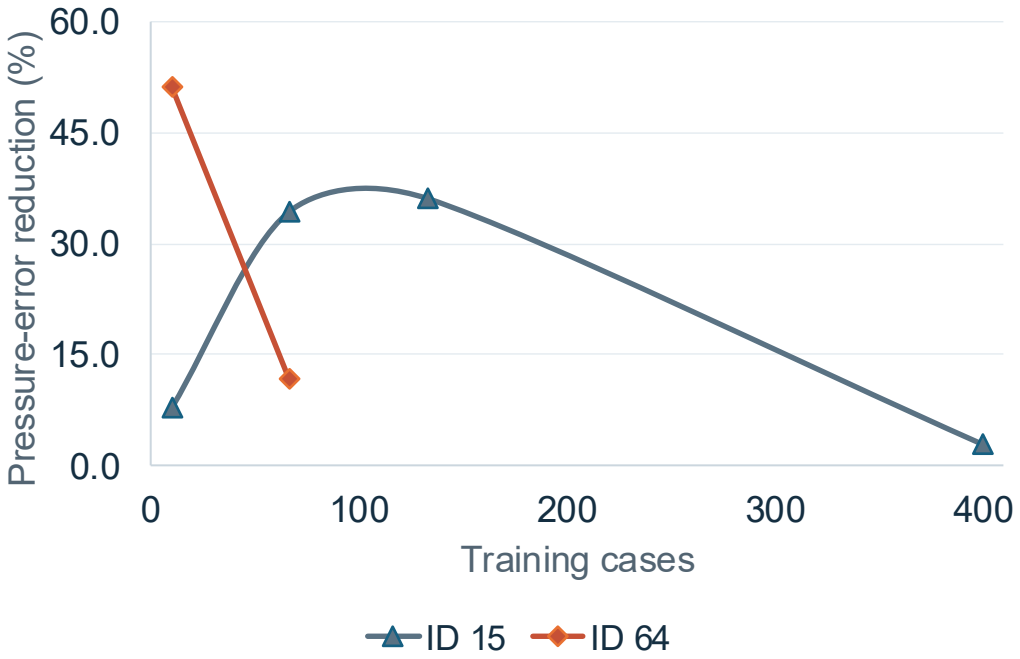
Full **400** Medium **133** Scarce **67** Super-scarce **11** training cases

Sensitivity to less training data



At 11 cases, pressure error rises 2.2–5.8× relative to each configuration's Full result.

Pretraining benefit versus scratch



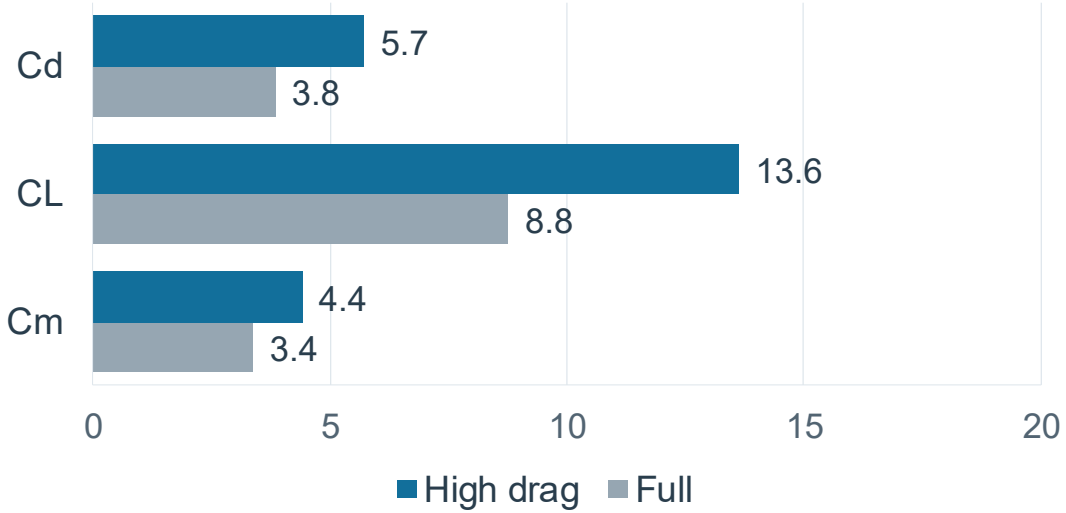
At 11 cases, pretraining cuts pressure error by 8% and 51% in the two studies.

OOD analysis: high drag

ID 37

Drag and lift errors rise more than surface errors

Force RMSE (counts)



Surface relative L_2 error (%)

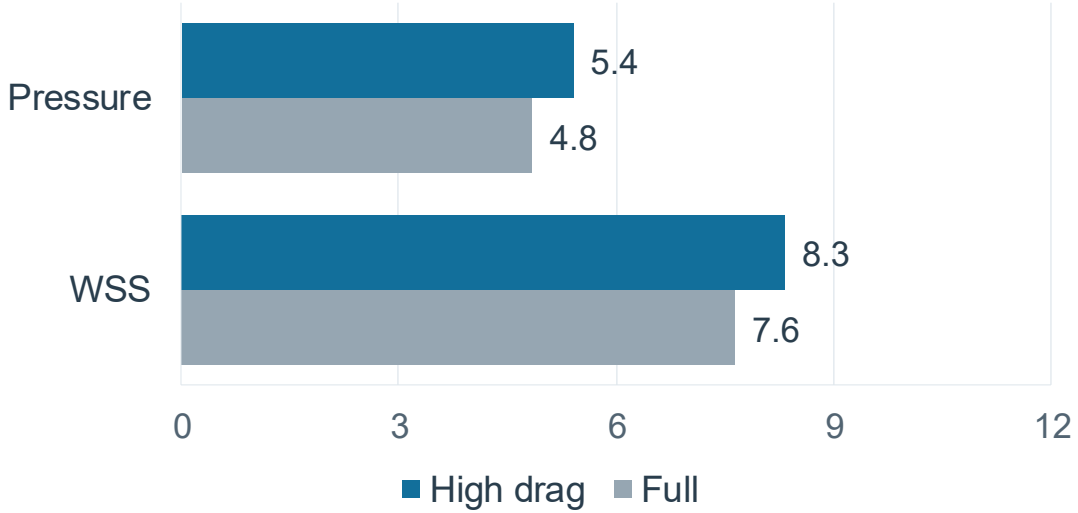


Table values: Full / High drag. Higher R^2 and lower RMSE or L_2 error are better.

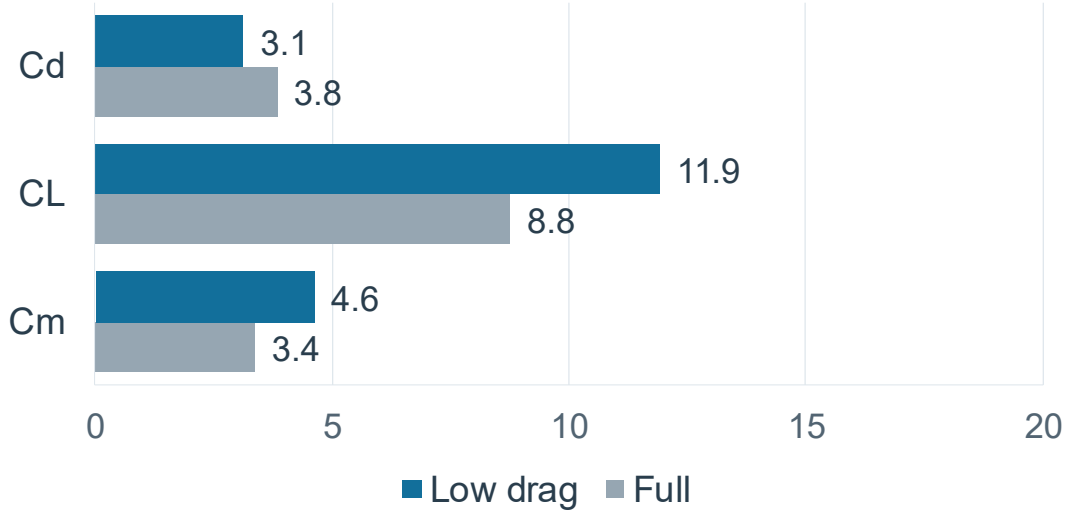
ID	Cd R^2	CL R^2	Cm R^2	Cp cuts R^2	Cp cuts RMSE
37	0.984 / 0.947	0.976 / 0.933	0.993 / 0.986	0.997 / 0.997	0.0141 / 0.0161

OOD analysis: low drag

ID 37:

Lower drag RMSE can coexist with lower R^2

Force RMSE (counts)



Surface relative L_2 error (%)

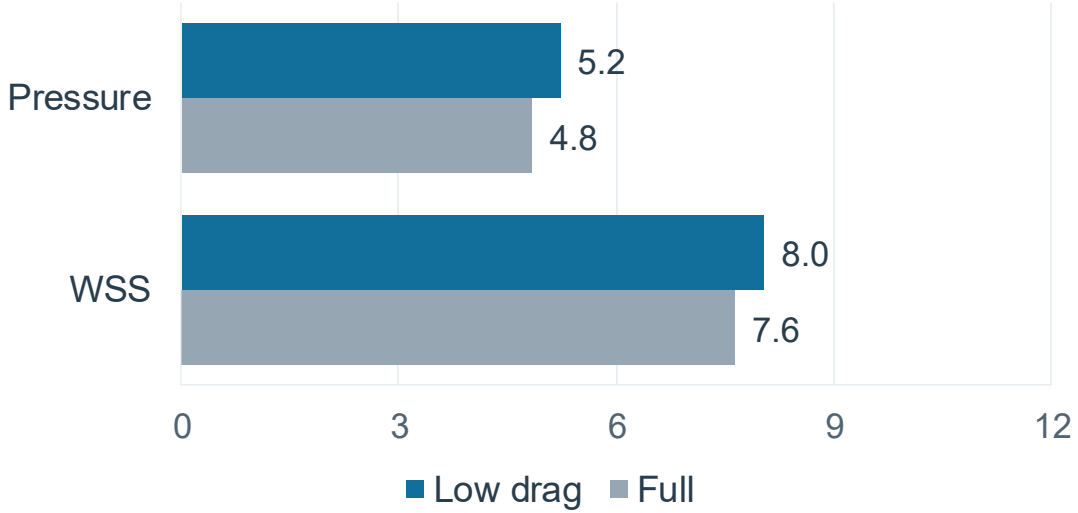


Table values: Full / Low drag. Higher R^2 and lower RMSE or L_2 error are better.

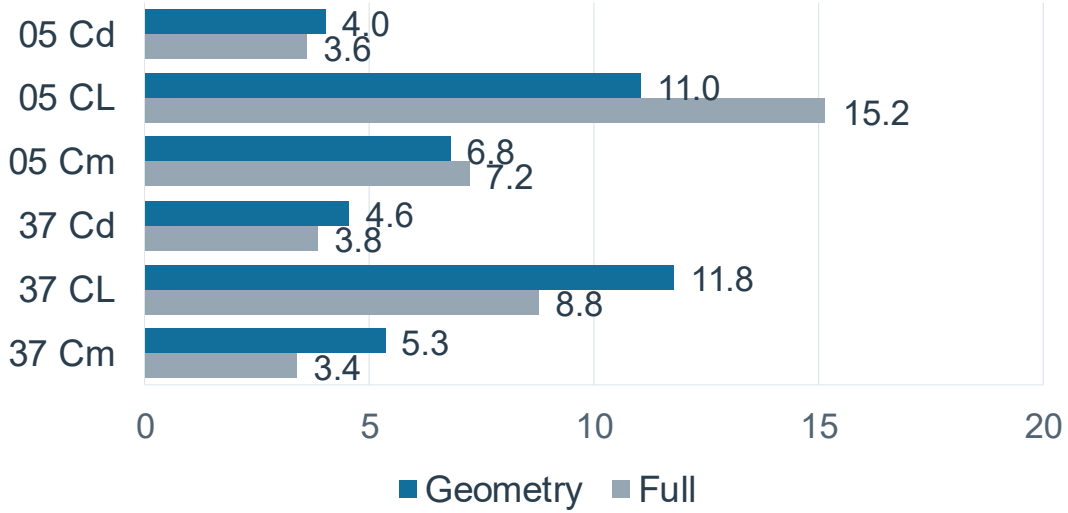
ID	Cd R^2	CL R^2	Cm R^2	Cp cuts R^2	Cp cuts RMSE
37	0.984 / 0.976	0.976 / 0.956	0.993 / 0.987	0.997 / 0.997	0.0141 / 0.0155

OOD analysis: geometry

ID 05: and. ID 37:

Both methods show larger surface errors

Force RMSE (counts)



Surface relative L_2 error (%)

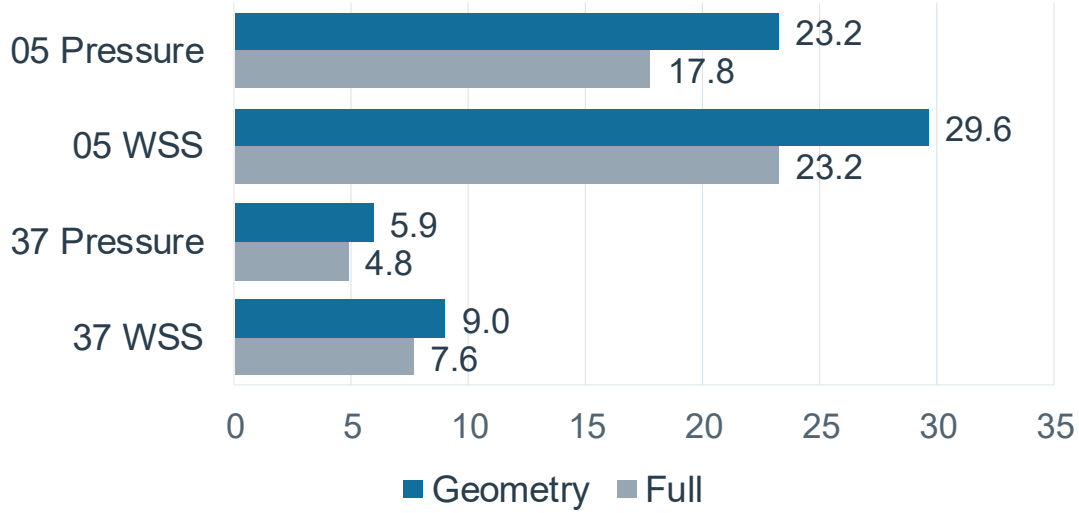


Table values: Full / Geometry. Higher R^2 and lower RMSE or L_2 error are better.

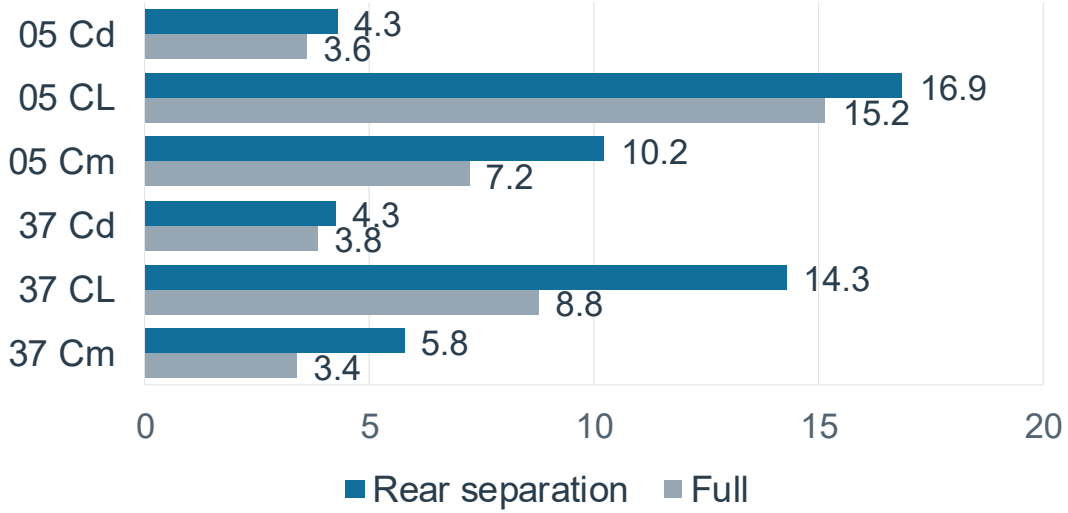
ID	Cd R^2	CL R^2	Cm R^2	Cp cuts R^2	Cp cuts RMSE
05	0.986 / 0.986	0.929 / 0.983	0.970 / 0.977	0.955 / 0.907	0.0599 / 0.0826
37	0.984 / 0.982	0.976 / 0.981	0.993 / 0.986	0.997 / 0.996	0.0141 / 0.0174

OOD analysis: rear separation

ID 05: and. ID 37:

Pitching moment is more sensitive than drag for both methods

Force RMSE (counts)



Surface relative L_2 error (%)

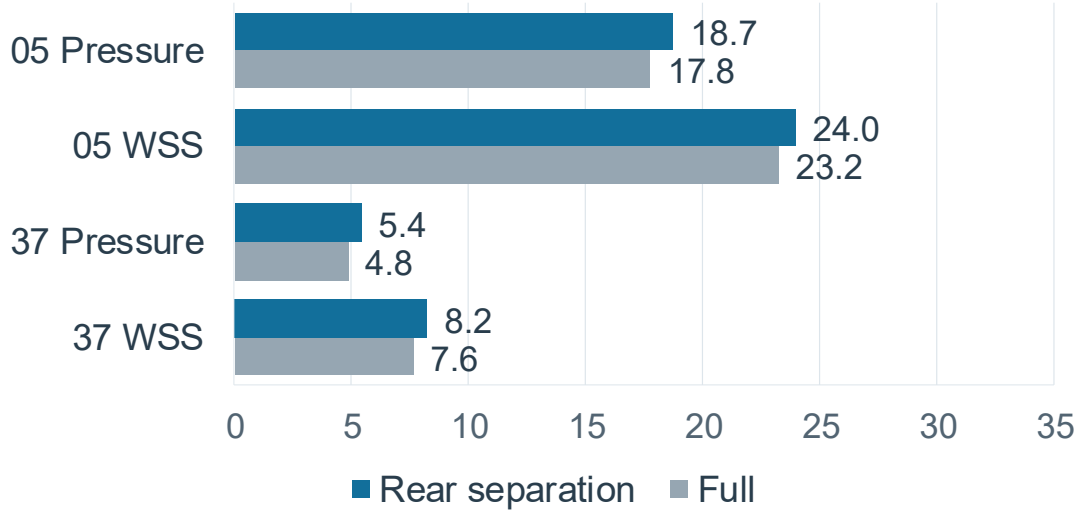


Table values: Full / Rear separation. Higher R^2 and lower RMSE or L_2 error are better.

ID	Cd R^2	CL R^2	Cm R^2	Cp cuts R^2	Cp cuts RMSE
05	0.986 / 0.980	0.929 / 0.901	0.970 / 0.939	0.955 / 0.949	0.0599 / 0.0628
37	0.984 / 0.981	0.976 / 0.929	0.993 / 0.981	0.997 / 0.997	0.0141 / 0.0157

Conclusions

Strong force predictions, with larger differences in flow-field accuracy

Force predictions are encouraging

Full split: drag RMSE is 1.7–8.1 counts. Models predict the correct drag-change direction for 93–98% of all geometry pairs. Small local design changes still need testing.

Flow-field detail remains important

Full split: pressure L_2 error spans 3.5–17.8% and WSS 5.6–23.2%.
Pressure cuts and velocity profiles reveal where the models differ.

More training data improves pressure accuracy

With 11 rather than 400 training cases, pressure error rises 2.2–5.8× in the available baseline comparisons, using the same 50 test geometries.

OOD sensitivity depends on the quantity

Only two IDs submitted OOD results, both surface only. Rear-separation moment RMSE rises 42–71%. Different training and test sets limit direct Full/OOD comparisons.

Next: broader OOD coverage and controlled geometry changes to test design reliability